*Frameworx Standard*

# Information Framework (SID)

*Common Business Entities – Location*

**Information Framework Suite**

**GB922 Location**

**Release 14.5.1**

**March 2015**

| | |
|---|---|
| **Latest Update: Frameworx Release 14.5** | **TM Forum Approved**<br>**Suitable for Conformance** |
| **Version 14.5.1** | **IPR Mode: RAND** |

# Notice

Direct inquiries to the TM Forum office:

240 Headquarters Plaza,
East Tower – 10th Floor,
Morristown, NJ 07960 USA
Tel No. +1 973 944 5100
Fax No. +1 973 944 5110
TM Forum Web Page: www.tmforum.org

# Table of Contents



# Table of Figures



# Business Entities

## Location Domain

### Disclaimer

This Location model is complex and has a number of subtleties. It provides a high level framework, explains broad concepts and gives illustrative examples. Actual solutions for particular requirements will be dependent factors such as

- country address standards
- state/ country / province map standards
- Service Provider addressing standards
- Equipment vendor equipment naming & addressing conventions
- whether textual only or graphical representations will be provided

It is expected that when this model is used in NGOSS Catalyst projects, the feedback will clarify its application & usage in particular situations.

There is obviously a limit to the amount of detail that can be included in a document like this. Those wishing to delve into more detail should consult the references used as the basis for this document.

## Introduction

At the most fundamental level [Zachman], [Coad-Archetypes] the world can be thought of as comprising of entities that answer the 5 fundamental questions.

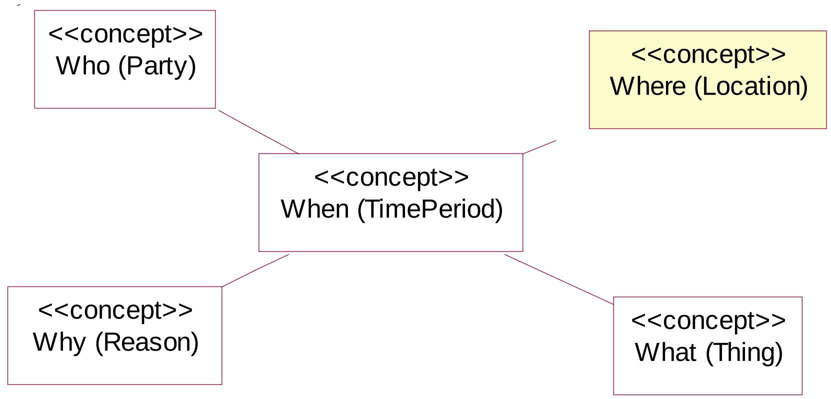



**Figure L.00 – Showing how Location Fits in**

This section of the SID answers the question “where ?” and deals with locations in the context of the eTOM, which looks at processes from a Service Provider’s point of view.

*“The eTOM is a business process model or framework that provides the enterprise processes required for a service provider.” [eTOM]*

A Service Provider will not be interested in planetary and galactic positioning, our interest in location will be earth centered.

We may not only be interested in places on or near the earth’s surface however. For example, we may also be interested in undersea locations of submarine cables and in the positions of communications satellites, and airplanes with mobile phone interfaces.

A Service Provider will be interested in the location of many things, and may ask:

- Where is the customer (more on this later)?
- Where is our equipment?
- Where is the fault and where is the nearest person who could repair the equipment?
- Where did the cyclone pass (and what equipment may have been affected)?
- Where is growth for this product?
- Where can we expand our business operations?
- Where are our service vans and the work sites (to optimize work dispatch)?
- Where (which shops) do we require products to be shipped?
- Where do I send the bill for Customer X's use of product Y?
- Where do I need additional employees?
- Who owns this property (to advise of digging)? {note property may be rented, i.e. customer living at property may not be owner}

Note that the business requirements will not just be Network related but will also be required by marketing, corporate strategic, etc processes.

In some cases these locations will be well defined points. In others they may be fuzzily defined regions of interest.

The Location model enables us to answer questions like "What is the nearest network access point to the Customer's location that can provide this product?" Network, Customer and Product Capability should not just be seen as separate location views, but as an integrated view.

## eTOM Requirements

While the eTOM has many **indirect** references that **may** imply location functionality (e.g. "product delivery") the only explicit location reference in the eTOM processes is in Workforce Strategy (HR – EM)

*"These processes create the strategies needed to ensure that the correct type, quantity and quality of staff will be available in right locations for future business." [eTOM]*

The only other eTOM references to location are in the figure “Services in the ICS Industry are Multiplying” which has an arrow marked “Location based services” and at the start of chapter 1 *“Everything, Everywhere, Everytime”*.

This model will provide a level of Location functionality that could be reasonably be expected to be required by the eTOM use cases.

## Customer Location / Location Based Services

In most cases, we will not know the actual location of a Customer (person). For example, even if we tracked the location of their mobile phone, they may have left it at home or lent it to a friend. Legal reasons may also restrict us from determining a Customer’s actual location (except in special circumstances – e.g. emergency calls).

From a Service Provider view we are probably asking “Where does the Customer require service?”

There are a set of services called “Location based services” [OLS]. These utilize the Customer Premises Equipment (CPE) location to provide additional Customer value. An example may be a service allowing a Customer to call using their mobile phone and find the closest “Pizza Barn”. The service could return the three closest Addresses as a SMS message or as WML or send a simple map on a 3G phone, showing the CPE location and the closest fast food locations. Note that this service would use the actual CPE location, not just the cell that the mobile phone is working off (see example 5).

## Graphical or Textual?

This section introduces the concept of spatial representation and algorithms. It is felt that explaining these concepts ahead of the model definitions will make the model easier to understand.

This facet of the SID model will provide for spatial (i.e. map display) manipulation of locations as this is widely used in operations centers, customer service and network planning areas of a Service Provider’s business.

From a service Provider’s point of view, GeoSpatial data can be categorized similar to the layers on a map, as shown in fig 1 below.

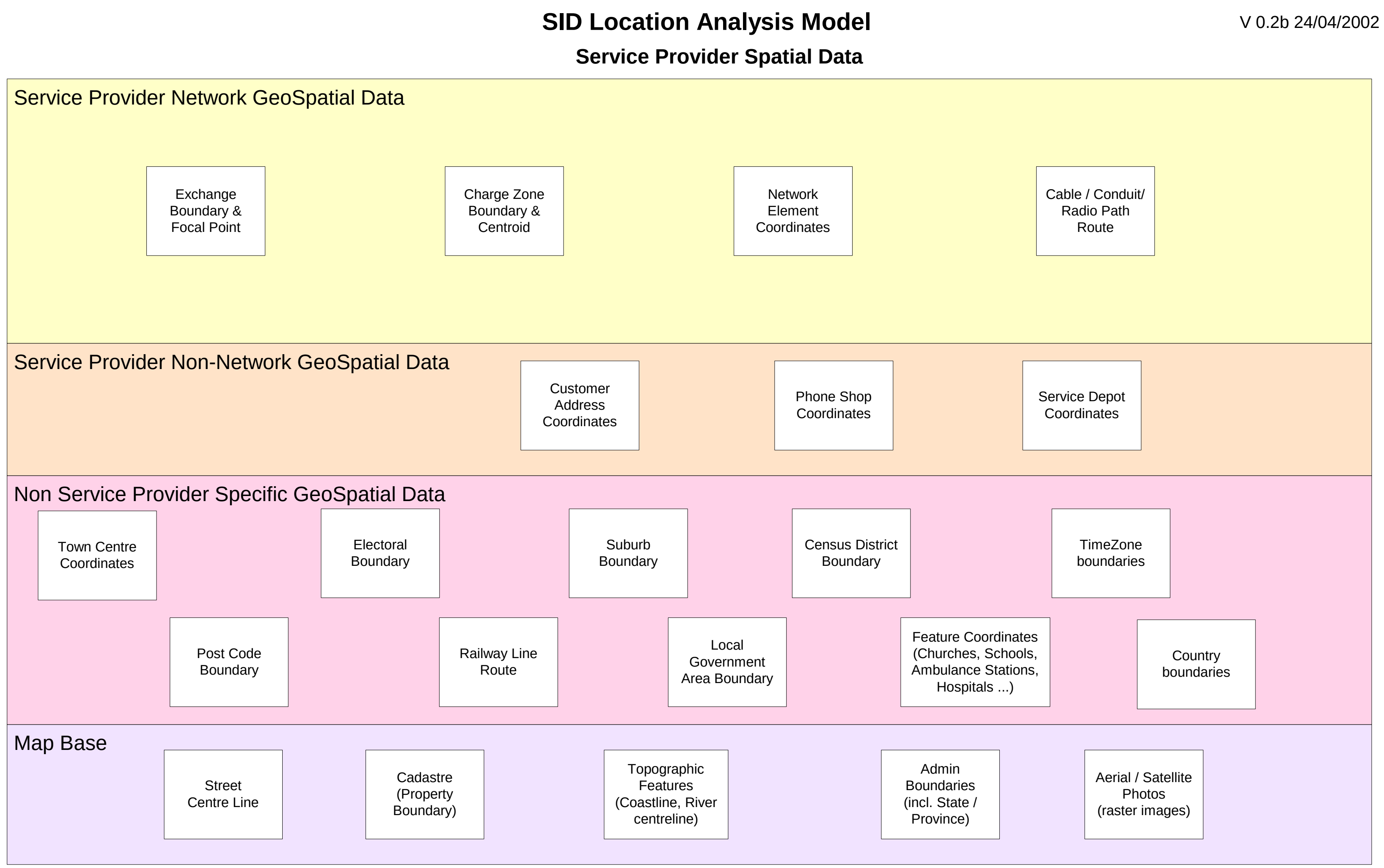



**Figure L.01 – Possible Service Provider Spatial Data**

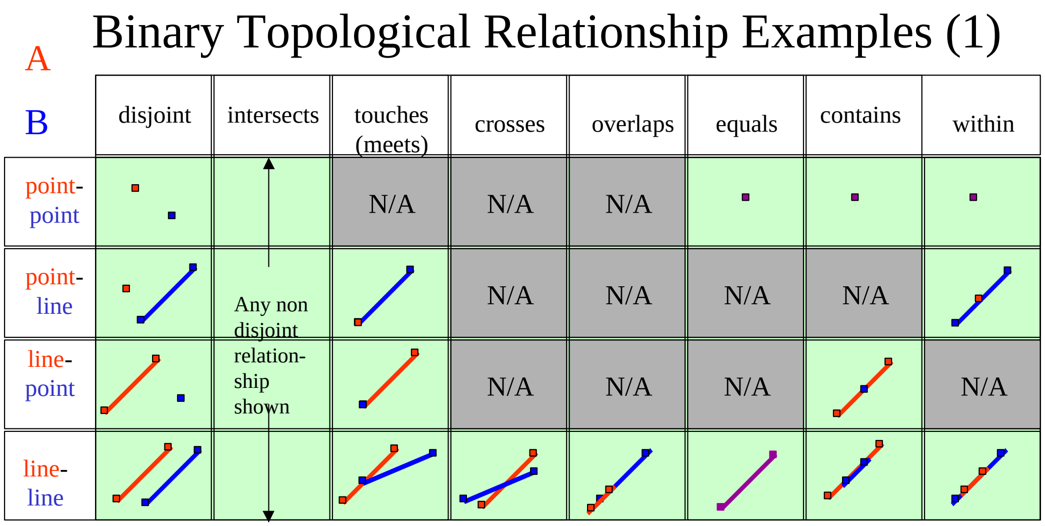
This model has been designed to be able to be used in textual only or textual/spatial situations, however it is recommended that spatial presentation is provided.

A common problem with a textual only environment is that spatial relationships are inappropriately approximated (e.g. hierarchies of regions are created, where the region boundaries don't match). [Hay]

Figs 2a and 2b show examples of binary spatial relationships for simple point, line and region geometries [OGIS], [Clementini], [Egenhofer], [JTS]. Readers who wish to learn more about this topic should consult these references.

If locations are to be related using associations, then the associations should match the spatial binary interactions (e.g. as an association type).




A & B disjoint - the boundaries & interiors of A & B do not intersect
A intersects B - A and B are not *disjoint*
A contains B - the interior & boundary of A is contained in the interior of B
A equals B - A and B have the same boundary and interior
A touches B - the boundaries of A & B intersect but the interiors do not
A within B - the interior of A is contained in the interior of B and their boundaries intersect
OR the interior & boundary of A is contained in the interior of B
A crosses B - For point-line, point-area, line-line and line-area. A is not *within* B and B is not *within* A and their intersection is of a lesser dimension than the maximum dimension of A,B
A overlaps B - A & B are of the same type (e.g. line-line) and A is not *within* B and B is not *within* A and their intersection is of the same type as A & B (also a line)

**Figure L.02a – Binary Topological Relationship Examples (1)**

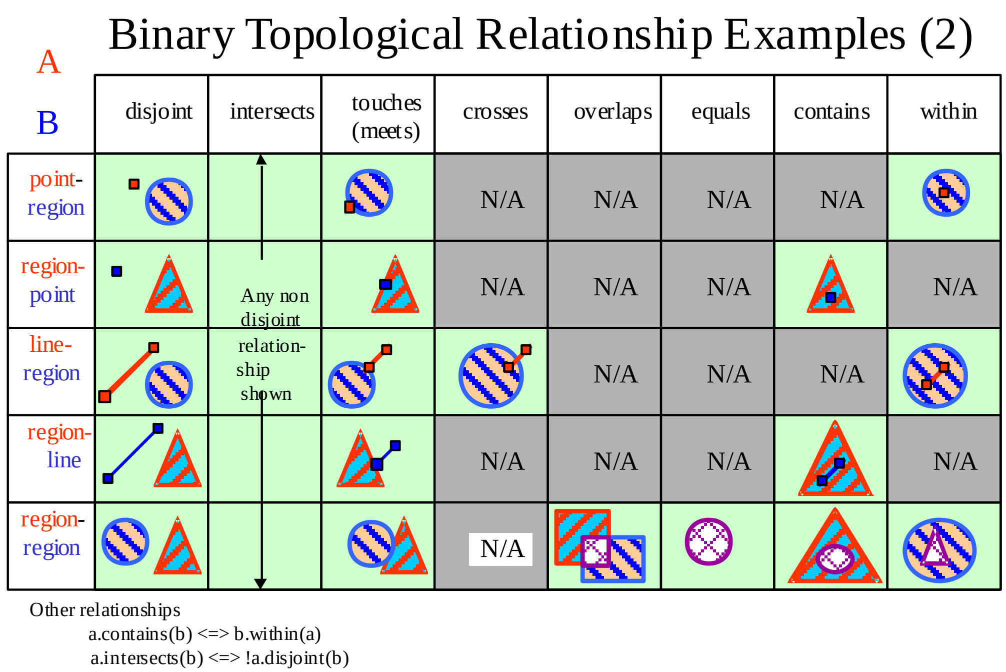



**Figure L.02b – Binary Topological Relationship Examples (2)**

We can see that the actual relationships are dependent on the types of geometries and that there are numerous variations, even with only two very simple objects.

This section has highlighted that spatial functionality is more than just a map pictorial presentation, but is fundamental to the data storage and how the data can be analyzed. This should be kept in mind while reading the rest of the document to determine whether a textual only representation can accurately support particular business requirements.

## Place

We will define Place as a generic concept that can answer the question “Where?”

We will define two types of Places:

- Geographic (External) – drawn on a map, defined using a geographic co-ordinate system (absolute world centered datum and projection), typically maintained using a GIS
- Local – drawn on floor-plans (suite / rack positions), defined using a Cartesian coordinate system relative to a local datum, typically maintained using CAD packages

Also see the section “Contrasting Geographic & Local places” for further discussion of this.

Places are an abstract modeling concept, used to generalize Location, Address and Site. In later sections we will explore these three different concepts. Figure 3 shows the main concepts and how they are related.

This model provides four options for locating something geographically:

- With a defined (named) location {sub class of Geographic Location}
- With an address of a location {Geographic Address}
- With a (named) site {Geographic Site}
- With a survey location (geocode) e.g. a GPS position for an offshore oil rig {Simple Geographic Location}

The way that the Place hierarchy is defined is the key to this model. The high level concepts are broken up into two levels of granularity, Geographic and Local, rather than by Address, Location & Site.

Note that various names for “Local” were considered. The following were considered (the reason for not using it is bracketed):

- internal place (is awkward when referring to things that are attached, not enclosed)

- relative place (confusing due to the use of relative geographic places)
- structural place (confusing if there is no structure)

So the term Local may not be perfect, but it is the best found to date.

Fig 2c shows how different types of Geographic and Local Places can be combined together.

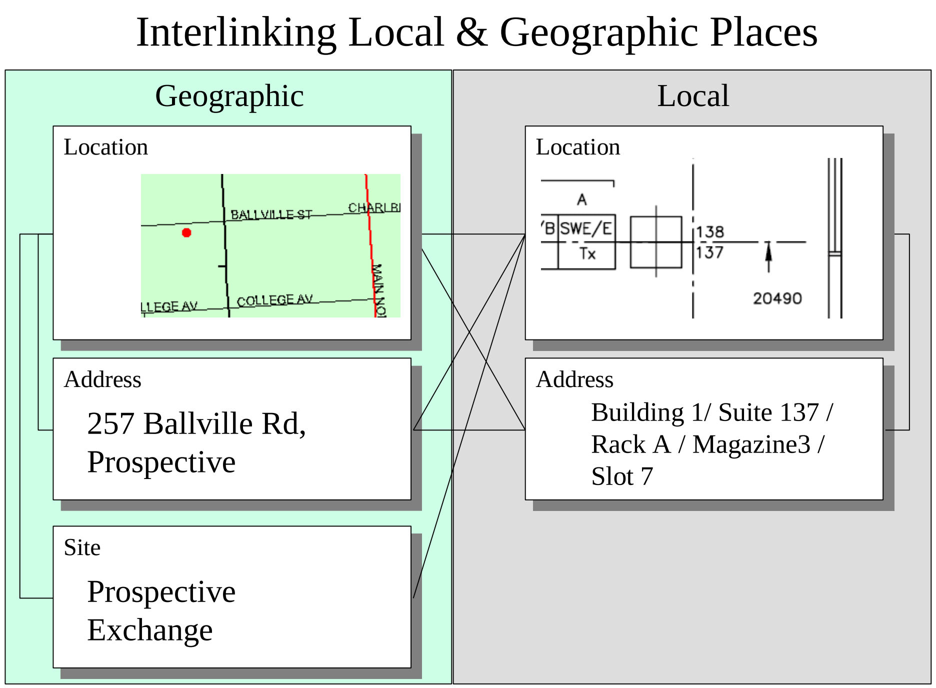



**Figure L.02c – Interlinking Geographic & Local Place**

If we need to communicate information about a Geographic Place, we can mark it on a map or give its address. If an address is given, then we will often need to look it up in a street directory (map).

We can specify a Local place as an Address or as a Location (by marking it on a drawing) and link it either to a Geographic Location, Address or Site.

Local Places are defined relative to a Geographic Place and we could choose as the Geographic reference point a Geographic Location, Address or Site.

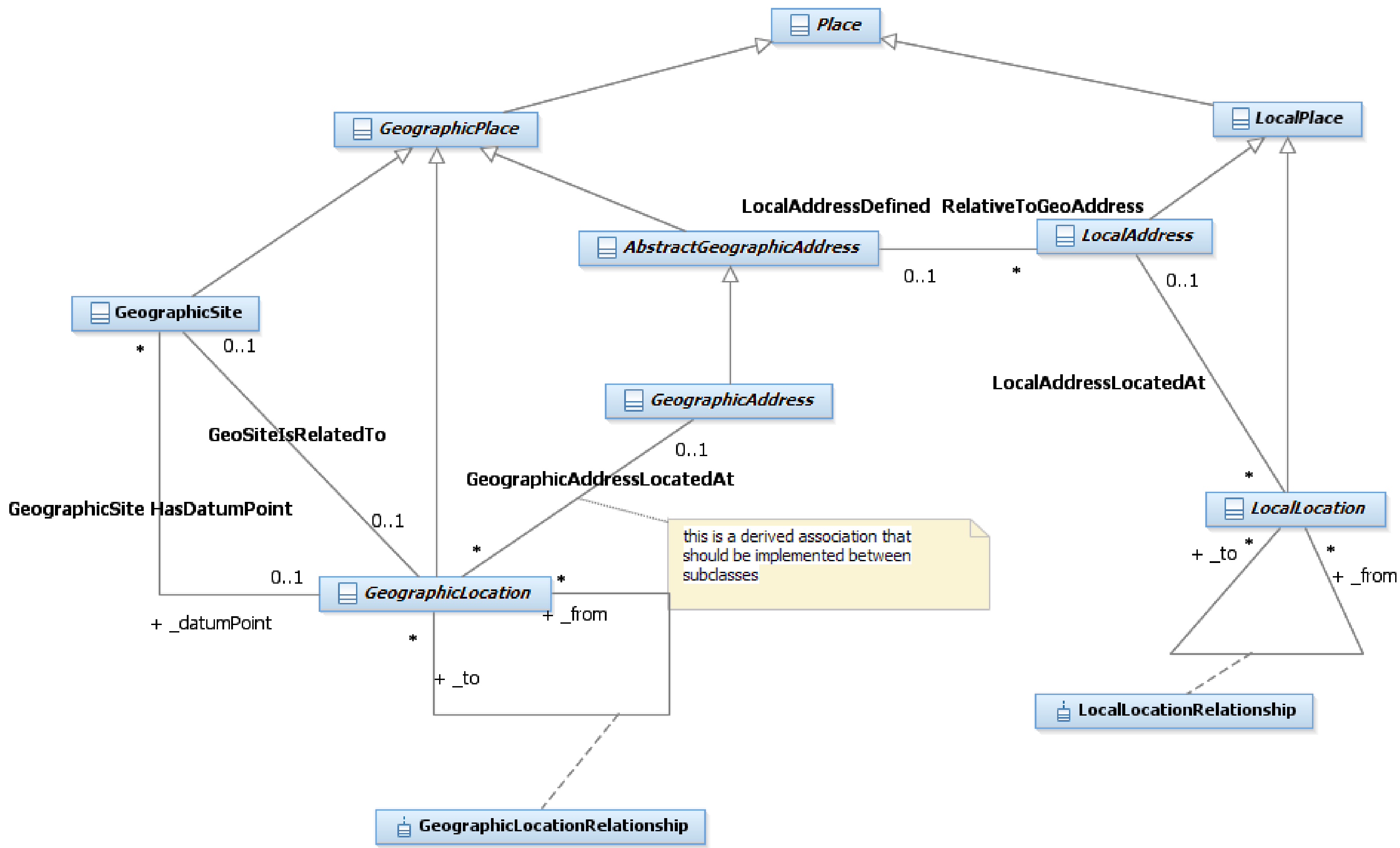



**Figure L.03 - Types of Places (Overview Hierarchy Diagram)**

If a spatial engine is not available, then it is likely that only Point geometries will be used (e.g. latitude, longitude, elevation) as this is the easiest geometry type to store textually.

## Location

This model defines Location as:

*“A defined place”*

A location usually has a unique code or name to allow textual identification.

Location is split into Geographic Location that relates to world centric places and Local Location, that relates to locally defined coordinate systems (usually defined relative to man-made features).

A Geographic Location has an optional Geometry attribute, allowing it to be displayed on a map.

As shown in fig 4 we will allow a Geographic Location to have many names. A name may change over time and could be defined in more than one language. Also a location may have more than one name in the same language, or a European and native name (Ayers Rock is also known as Uluru).

*“A name is an informal way of identifying an entity” [Fowler-AP].*

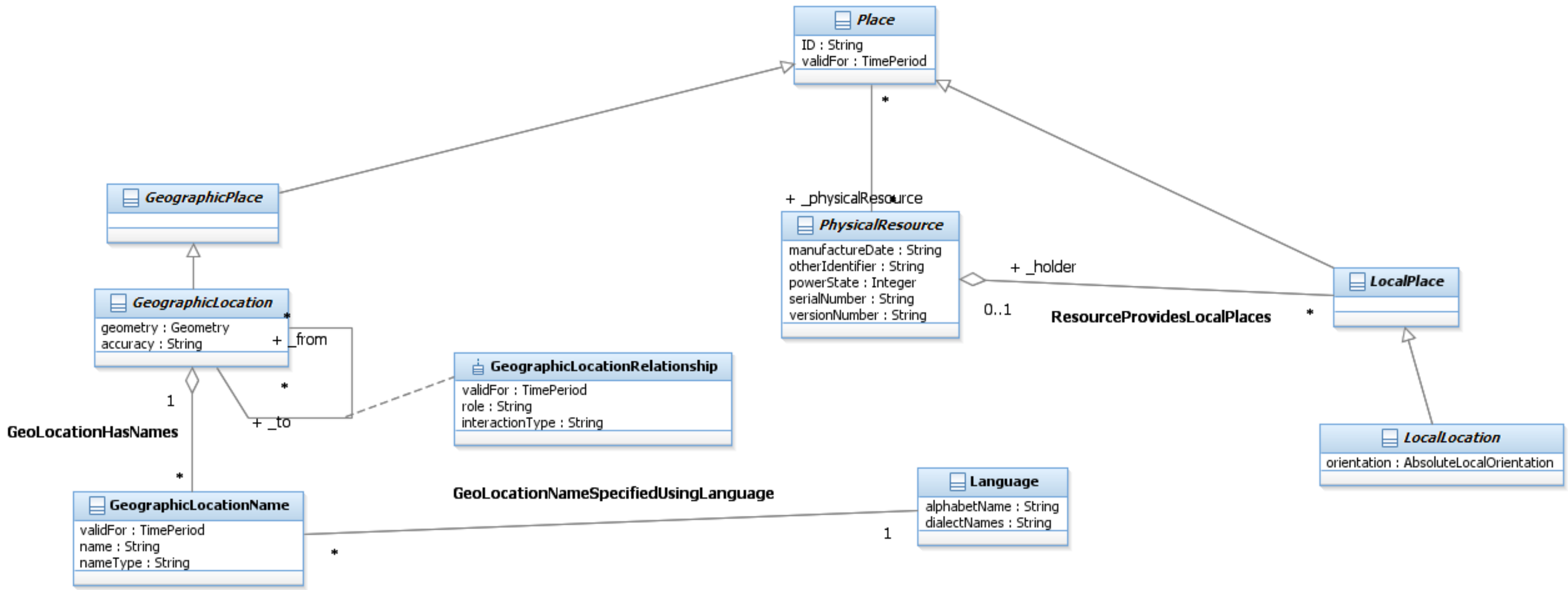



**Figure L.04 - Location**

## Address

This model defines Address as:

*"An address is a structured textual way of describing how to find a Location. It is usually composed of an ordered list of Location names based on context specific rules."*

Address is split up into Geographic Address and Local Address. Geographic Addresses describe Geographic Locations and Local Addresses describe Local Locations.

Geographic Address caters for Property and (physical) postal Addresses.

Geographic Sub Address caters for Property Sub Addresses.

Local Address caters for a level of granularity smaller than Geographic Address

While it may be possible to derive a Geographic Address from a map point using Geographic Location names [Hay], there are advantages in storing Geographic Address as a separate entity.

Geographic Addresses may be required:

- if the Address cannot always be calculated reliably from a map (e.g. to Post Office standards)
- if a spatial engine is not available
- if suitable spatial street, suburb, post code data is not available
- for logical addresses (e.g. PO Box)
- to achieve required response times. It takes a lot of computation & spatial facilities to derive the address from a map position, so storing it redundantly as text makes sense (easy for selection etc.)

Country specific standards should be implemented using Policies, Templates & Rules.

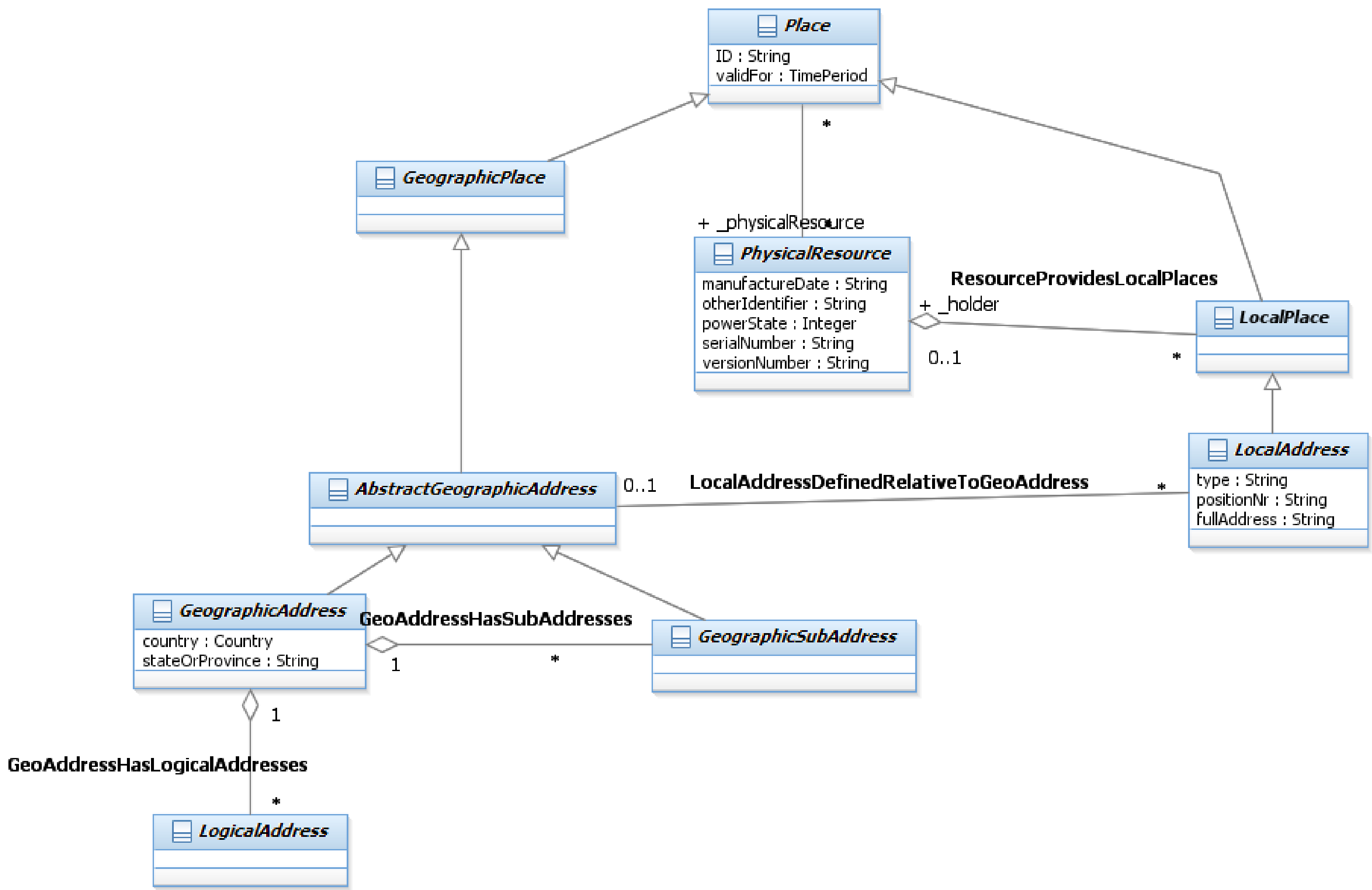



**Figure L.05 – Address**

## Logical Address

The term Address is often used to describe:

- Addresses that enable you to find a location (e.g. Street Address), which we will define as a physical address
- Addresses that don't (e.g. PO Box address, email Address), which we will define as a logical address

A logical address allows us to send something (tangible or intangible) to a central point where it can be retrieved. With email, a message is sent to the recipient's ISP mail server. The recipient can then connect to the server from anywhere in the world and retrieve the message. The intermediate location is not of interest to the message sender.

Only logical addresses that relate to Location (e.g. Po Box Address), will be covered in this model (see SID Party Domain model / Contact Medium for e-mail addresses etc.).

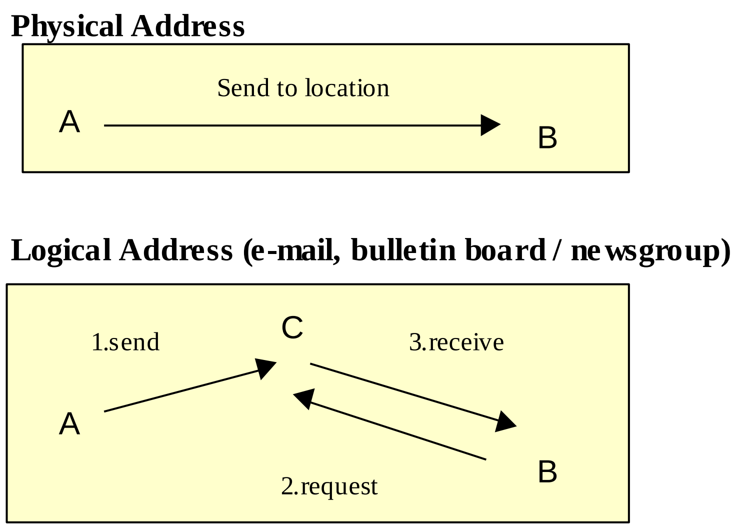



**Figure L.05a – Logical & Physical Addresses**

Note that Network Addresses (e.g. IP Address, Phone Number) are not Location related (as per our definition) and hence are not covered in this model. They will be covered in a later version of either the SID Service or Resource model.

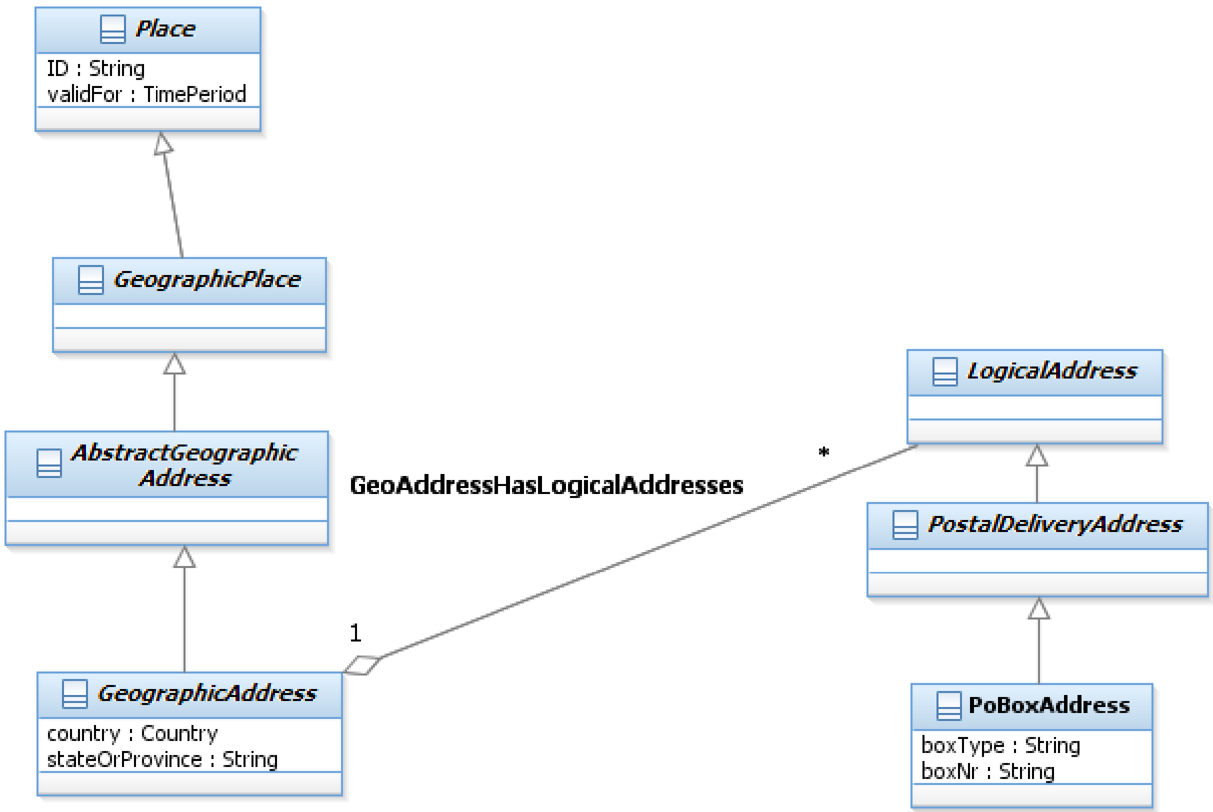



**Figure L.06 – Logical Addresses**

### Site & Site Role

This model defines a Site as:

*"A place that has (existing, planned or former) physical things or characteristics of interest to the Service Provider. Things of interest include Product Instance(s, Service Instance(s)) and Physical Resource Instance(s). Characteristics of interest include associated Parties (property owner, property lessee)"*

Site provides a convenient linking point to other parts of the SID model.

A Site may be assigned many roles (or functions).

Site Role types should be restricted to functions that cannot be easily derived from entities linked to the site.

Some examples of Site Roles are:

- Asset Location
- Transmission Access Point
- Competitor Access Point
- Customer Access Point
- Equipment Delivery Point
- Cultural Site
- Protected Area
- Exchange Site

Site Roles should NOT be used for:

- Network Equipment Types (e.g. ADSL site)
- Network Equipment Roles (e.g. CCS7 Node – this is a role of the Resource)

Later in the document we will also define Geographic Location and Geographic Address roles. These other roles are modeled differently, which is why there isn't a generic "Place Role" entity defined.

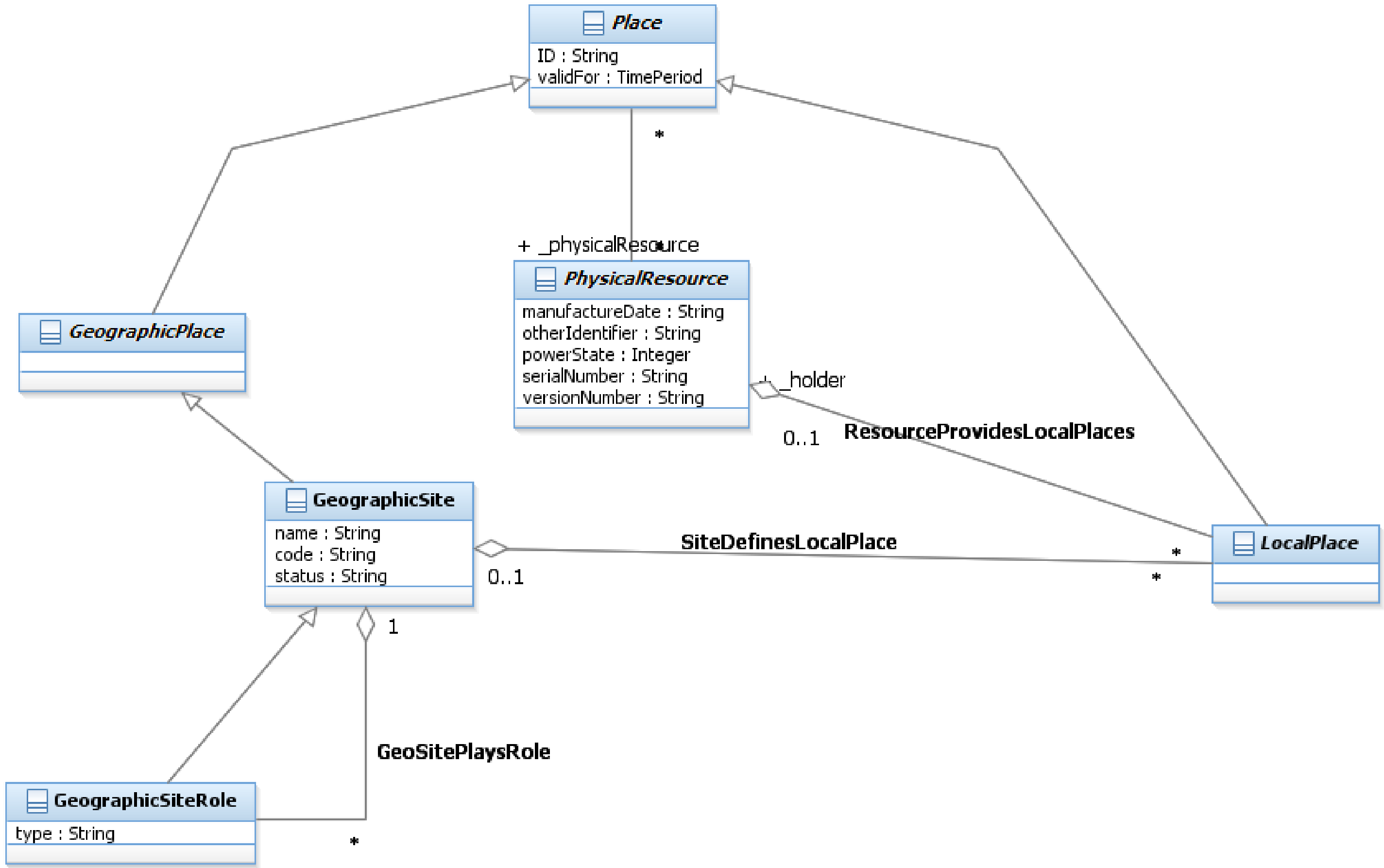



Figure L.07 - Site

## Place, Location, Address & Site

Figure 8 summarizes the main entities shown in figures 4-7

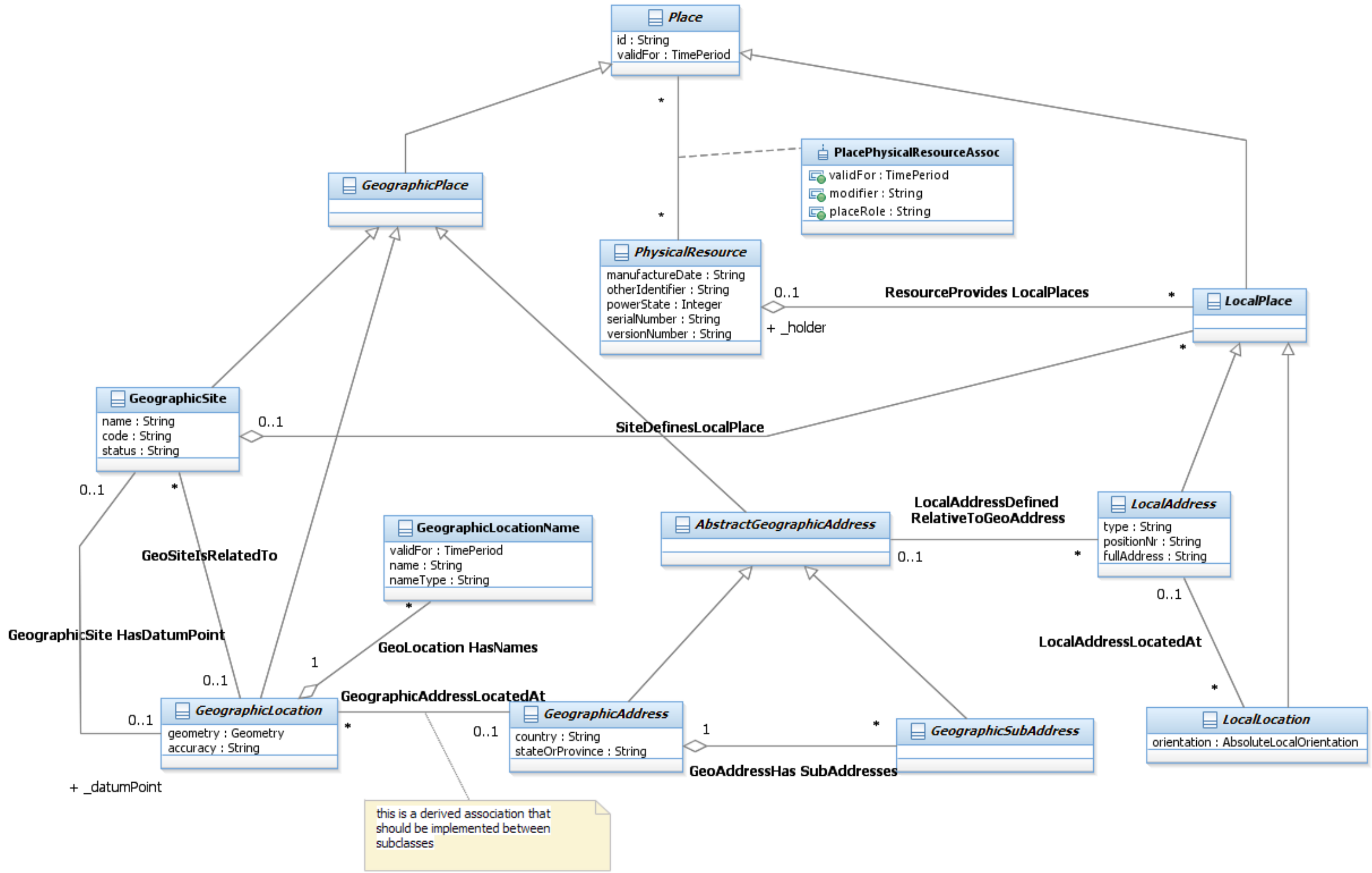



**Figure L.08 – Place, Geographic Location, Address & Site**

This model allows for changes in location detail over time. A Service Provider may start by having external address or Geographic Location information. Then other information, including Local information, can be added later as required (refer example 1).

## Geographic Geometry

This model adopts the Open GIS Consortium Geometry model [OGIS].

Whilst in theory any spatial engine / model could be used, use of a standard interface allows for component interworking.

Refer to document "OpenGIS® Simple Features Specification for SQL" at http://www.opengis.org/techno/implementation.htm for an explanation of the model. Note that the Geometry classes are not documented in the "data dictionary" section at the back of this document.

For messaging and data interchange of Geographic information, the GML standard is recommended.

*"The Geography Markup Language (GML) is an XML encoding for the transport and storage of geographic information, including both the spatial and non-spatial properties of geographic features." [GML]*

The Geometry classes should be considered as a closed set, with Geometry as the root class and no subclassing of any of the classes shown in fig 9. This aligns with the proposed ISO standard [ISO / DIS 19109].

If a spatial engine is not available, but geocodes are required, then a Point entity could be substituted.

The SID model uses Geometry as an attribute of the Geographic Location entity.

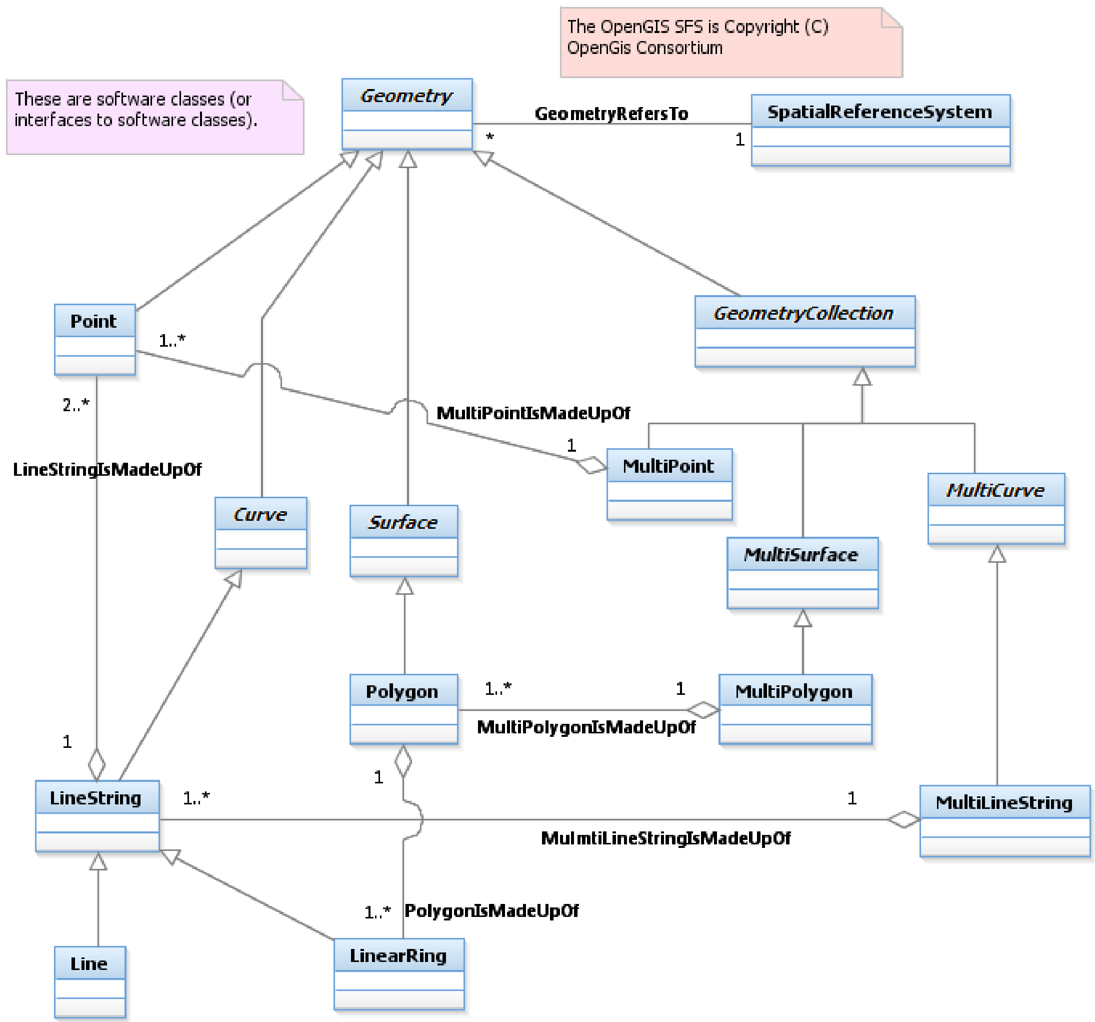



**Figure L.09 – Geographic Geometry [OGIS]**

Astute readers will have noticed that the OGIS Geometry class implements the Binary Topological Relationships shown in figs 2a & 2b.

## Geographic Locations

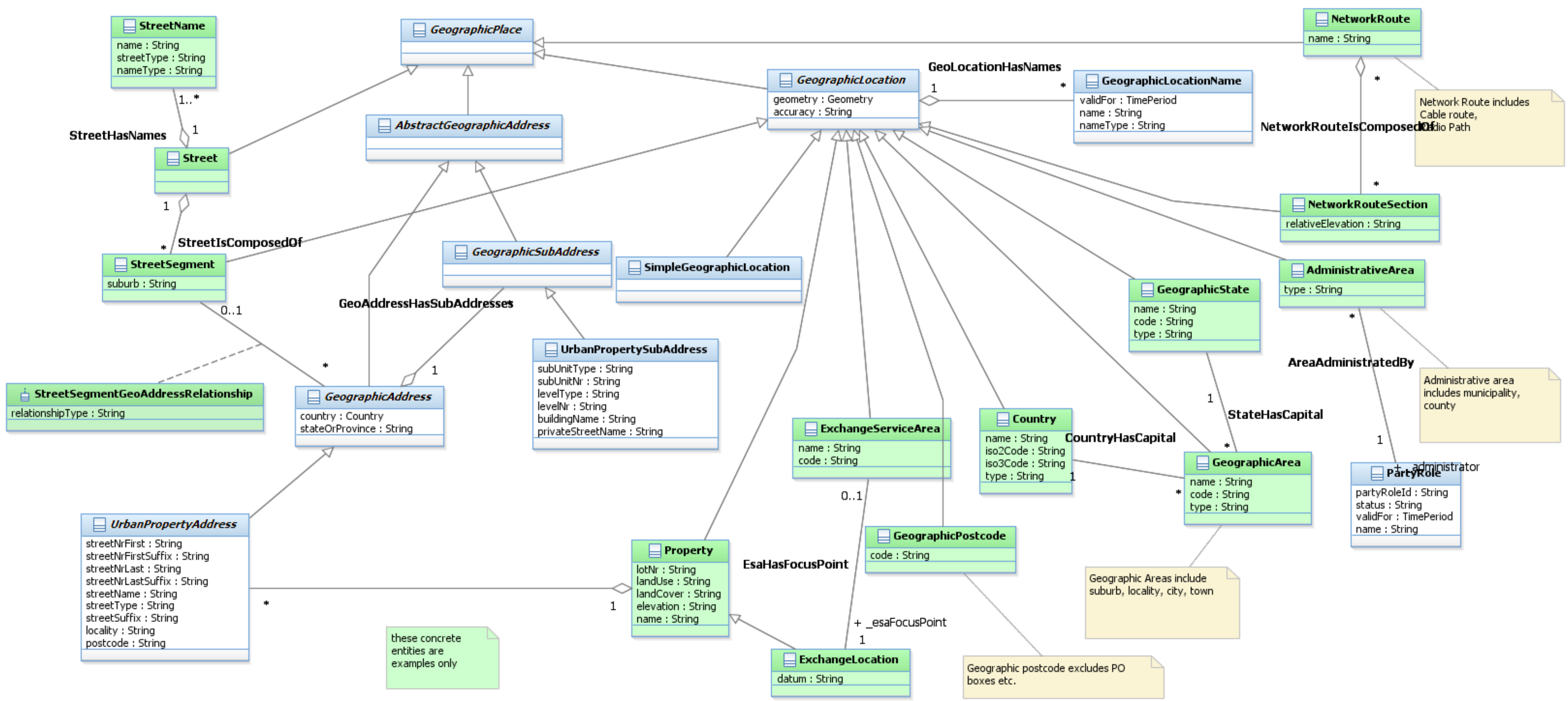



**Figure L.10a – Examples of Possible Geographic Locations**

Figure 10 shows as an example, an Exchange Service Area, which models an Exchange (Central Office) and the area that it serves. Other examples of Service Areas include Digital Radio Concentrator Systems (DRCS), Fixed Radio Access (FRA) service areas & Pay TV service areas. For wireless service areas (e.g. WiFi, 3G mobile), it may be useful to model the signal strength contours as a collection of areas.

A Simple Geographic Location entity should only be used (fig 3) when there is no unique textual identifier (name or code).

Note that there is a Spatial Data Standard [SDSFIE/FMSFIE] that defines around 1,500 feature types. It is not recommended that NGOSS uses its data model, but NGOSS should try and align with their attributes (feature type, name, zymology). It would also be useful if the SDSFIE/FMSFIE standard defined a set of XML schemas that could be used for data interchange (perhaps utilizing the GML [GML] standard).

To display features on a map we need to define the display format. Fig 10b shows how each type of Geographic Location can be assigned a symbology definition.

Figure 10 has two options for naming of GeographicLocations. If a GeographicLocation type only has a single name, then the name should be included as a simple attribute (e.g. Country). If the GeographicLocation type is likely to have names that change over time, or has alternative names (e.g. native names), then the GeographicLocartionName entity should be used (e.g. Administrative Area).

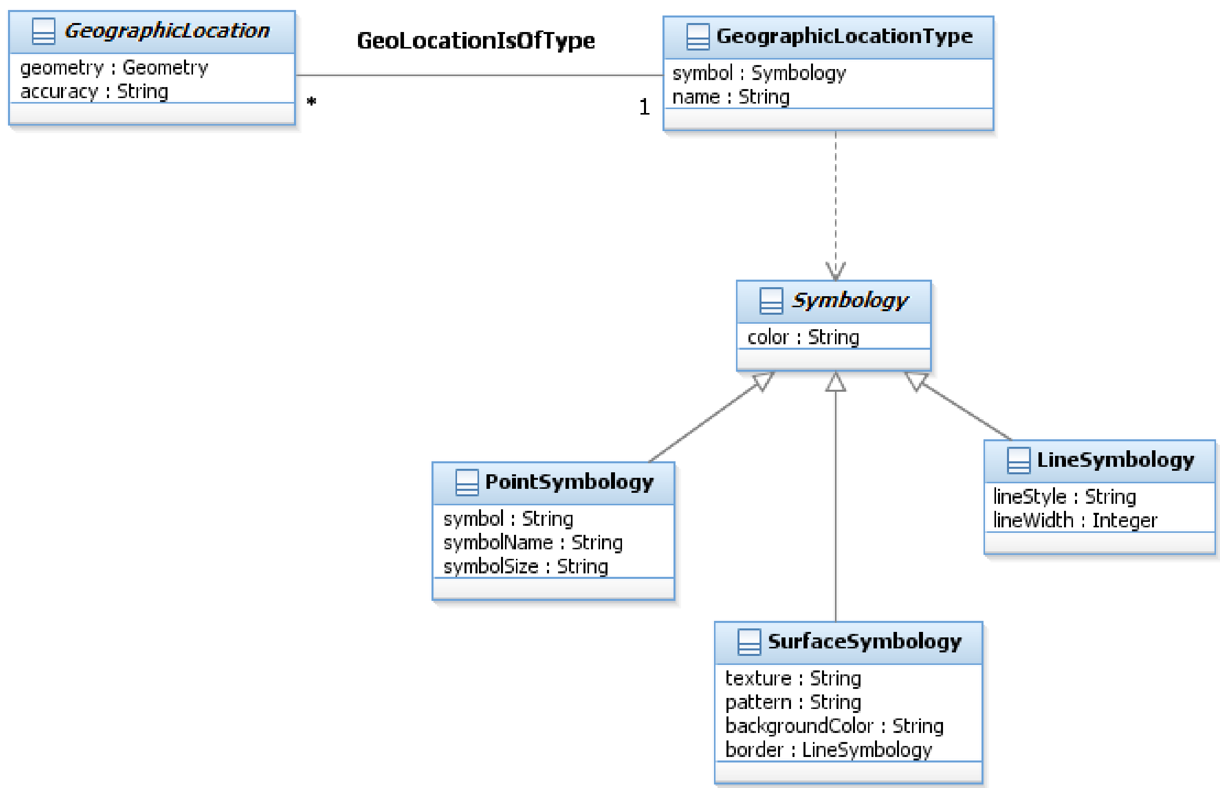



**Figure L.10b – Geographic Location Symbology**

## Contrasting Geographic & Local Places

We will now look at the contrasts between Geographic & Local Places.

In map representations, many of the features that we display are unique. For example

- country boundaries
- municipal boundaries
- railway lines
- property boundaries

are influenced by natural features (coastlines, hills, rivers etc.).

In the Local (artificial) world, many of the features that we wish to represent are made to specification.

On a map we are usually interested in where something is (e.g. where is the hospital?) and want to limit the detail so that the result is not too cluttered.

For Local places we are usually interested in both the position (placement) and orientation of features (where is the modem cabinet and which way does the front face?) as shown in fig. 11c.

On a map we use a two dimensional planar representation as we wish to view large numbers of features at a low level of detail.

For Local places, three dimensional modeling is required to accurately represent:

- multistory buildings

- rack layouts
- card positions
- things mounted on walls
- obstructions like low ceilings, windows, air conditioning vents

So we can see that we have different needs between Geographic and Local Places and that our model representation needs to reflect this.

## Local Places

[Oracle Spatial Concepts] defines Local Coordinate Systems (Non Earth) as:

> *"Local coordinates are Cartesian coordinates in a non-Earth (non-georeferenced) coordinate system.*
> *…*
> *Local coordinate systems are often used in CAD systems, and they can also be used in local surveys where the relationship between the surveyed site and the rest of the world is not important."*

Local Places are provided by a type of Physical Resource that we shall call an *EquipmentHolder*. An EquipmentHolder could be an *Enclosure*, that encloses the places inside it or a *Structure* that provides support for things to be attached. EquipmentHolders can be inside other EquipmentHolders, forming a hierarchy.

Structures can be attached to other structures, also forming a hierarchy (where the structure providing the support is the parent).

Structures can be contained in EquipmentHolders and EquipmentHolders can be attached to Structures, forming a mixed hierarchy (e.g. a single channel radio transceiver subrack is installed in a weatherproof box which is attached to a tower).

Local Places are defined relative to their EquipmentHolder.

A rack/ card position can have a status of empty or filled and this can change over time.

> e.g. an EquipmentHolder is installed with a card position empty. A card is later installed, and the card position is filled. The card fails and is removed and sent off for repairs and a replacement card is installed in the same position.

The SID Physical Resource model keeps track of the card; the SID Location model keeps track of the position.

We will define a *Housing* as an Enclosure that is not contained in anything (i.e. the outermost level of containment).

We will define a primary structure as a free standing structure (i.e. the outermost level of attachment).

Housings and Primary Structures may be related to a geographic location (i.e. be located on a map).

Some common Housing types are building, equipment cabinet, street cabinet, pit (underground utility box), housing, rack, sub-rack.

Some common Structure types are tower, pole, power pole, radio mast. A radio tower may have a number of rails with positions that can be used to attach antennas and dishes. Each of these rails could also be modeled as a structure.

Local Places are dependent on the SID Physical Resource model definitions. The Physical Resource model will only model to the “field replaceable unit (FRU)” level (i.e. it will not model individual resistors and ICs).

Local Places can be categorized in various ways. Some things to consider are:

- Above ground (mast), on ground (building, street cabinet) or underground (pit)
- Containment (inside) versus attachment (on)
- EquipmentHolders, which are contained in something (rack), versus those that aren’t (street cabinet). A street cabinet could be thought of as a weatherproof rack that also secures the equipment (from theft).
- Logical versus physical Holders (e.g. suite versus rack)
- Ownership. A housing may or may not be owned by the Service Provider (e.g. Leased building, shared site with leased space)

An EquipmentHolder may provide:

- “General space/ free form space” (e.g. floor space in a building – can be used for almost anything)
- “Managed / restricted / pre-defined space” (e.g. card position – restrictions on what can be plugged in)

A managed space EquipmentHolder entity, on instantiation, should create its Local places e.g. CMUX universal card (SPEC 227L - width & height, pin-out)

We will use a Cartesian local coordinate system for Local Locations. The Local coordinate system need not be north-south or earth surface aligned. This means that the local coordinate system definition will need to document a position & orientation of up to six degrees of freedom. In most cases the Holder or Structure will be aligned to the earth’s surface (i.e. not tilting like the tower of Pisa) and we will only require a position & orientation 4 degrees of freedom as shown in fig 11a.

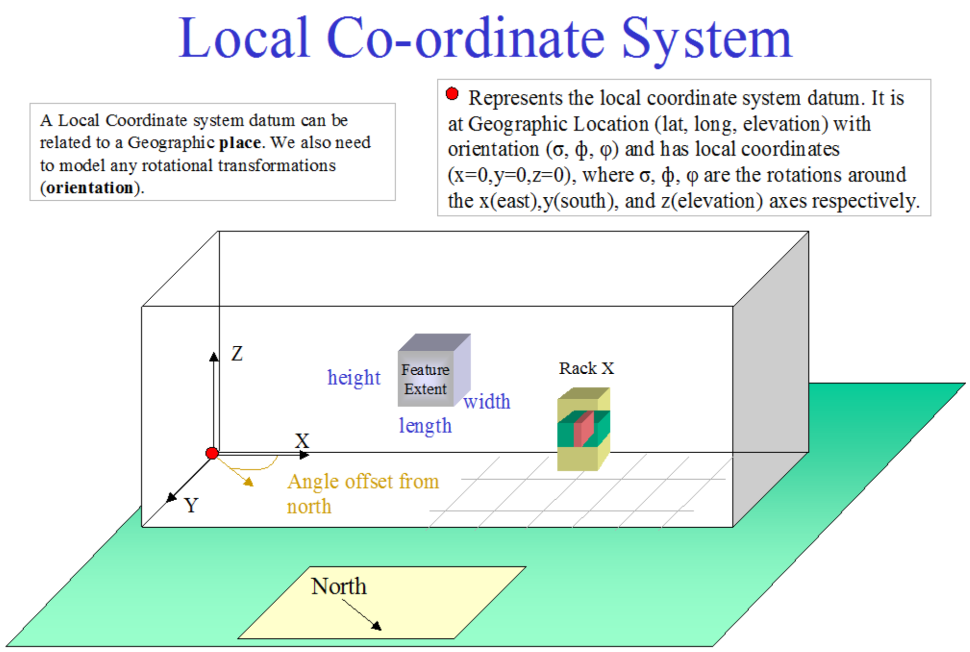



**Figure L.11a – Local Coordinate System**

Physical Resource specifications (in the SID Physical Resource model) should include three dimensional geometric representation(s). The geometric representation should include a minimum bounding volume and also allow for two dimensional representations (each face of the minimum bounding volume) and provide reference point(s) for the shape (for placement).

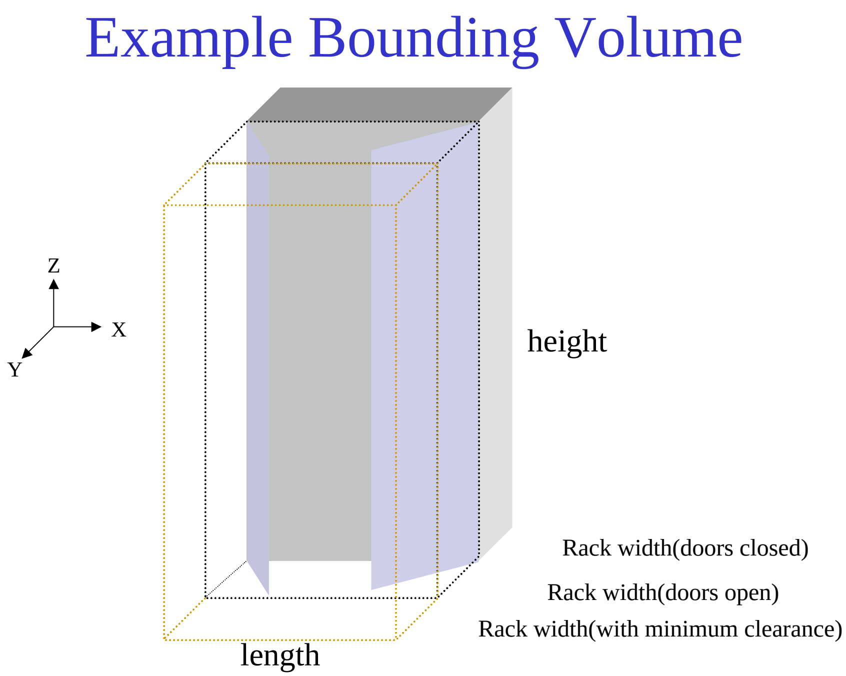



**Figure L.11b – Example Bounding Volume**

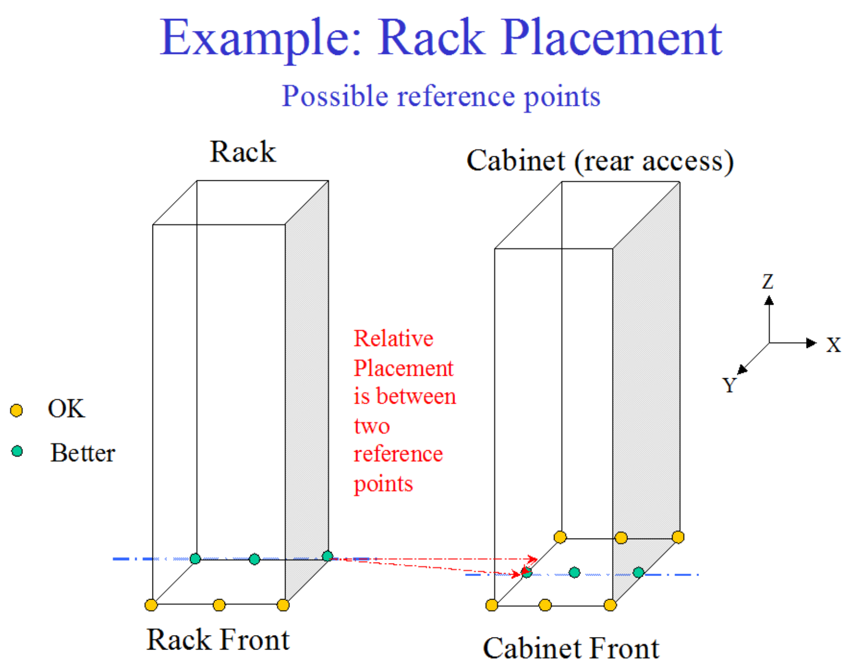



**Figure L.11c – Example of Rack Placement**

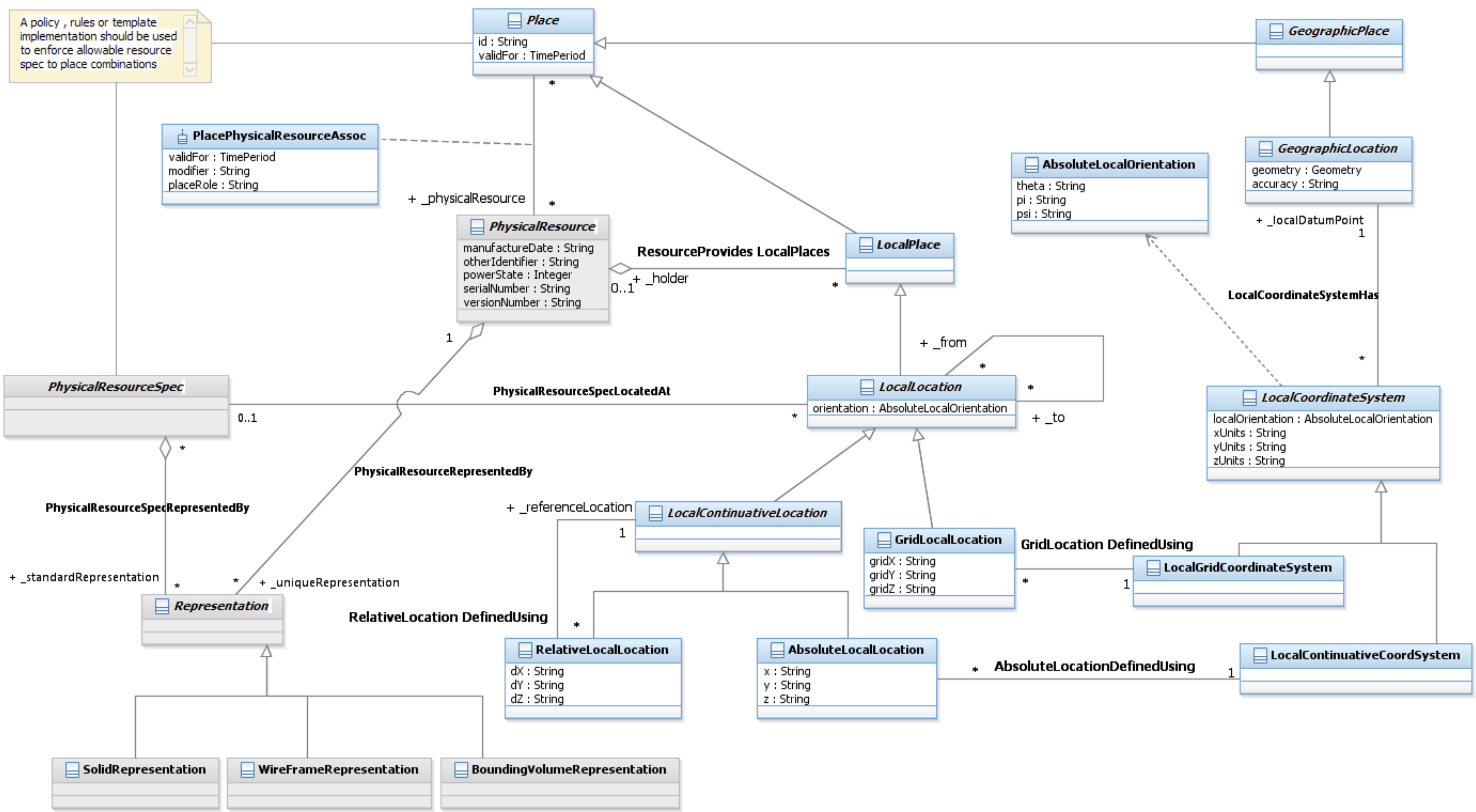



**Figure L.11d – Local Places are Defined using Local Coordinate Systems**

Note that a Grid Layout would be especially useful for rooms with raised floors where cabinets are installed in floor tile positions, card positions in a subrack, or subrack positions in a rack with rack unit connection points.

Note that we allow for both Physical Resources and Physical Resource Specifications to have representations. It would be expected that only unique physical resources (e.g. buildings) would need individual representations.

Note that in figure 11, we don't enforce the concepts of holders and housings in the static model. This is because the concepts are dependent on the status of the static resource. For instance, it may appear that a card can only exist in a card position (in a sub-rack etc.). This may be correct for a card with a status of "installed", but if the status is "spare" then it may exist in a store or if the status is "faulty-returned" then it may exist at a manufacturer's repair center.

### Types of Geographic Address (and Sub Address)

Addresses may have aliases, such as corner properties having more than one street frontage, or properties with rear entrances. We will require, however, that one of the addresses is chosen to be the Principal Address and the others are marked as aliases. Also roads and localities may have name aliases that will lead to address aliases.

We also have "vanity address aliases", where people use a different address to appear to live in a classier suburb. These will be modeled as Address aliases.

A location may have many addresses. An address may also be used as a location name if no distinctive location name (e.g. Buckingham Palace, The Pentagon) exists.

As well as simple "household" addresses, we need to be able to handle complex Addresses, which exist where a Property has multiple owners, tenants, access points or facility points. These are handled using Geographic SubAddress.

Examples of places that have complex addresses are:

- private streets
- caravan parks
- universities and other large educational institutions
- hospitals
- military bases
- retirement villages and other community developments
- entertainment / recreational parks

- marinas
- (marine) harbors / ports
- airports
- resorts
- shopping centers
- sporting venues
- townhouse developments
- industrial complexes, business/ commercial parks

It is likely that we will require different Geographic address rules for each country. This is represented by the entity Address Policy and the subclassing of Geographic Address in fig 12. Note that the address rules could be implemented using templates, or policy or rules engines.

A SubAddress is used for addressing within a property. It may refer to a building, a building cluster, or a floor of a multistory building.

e.g. a property with address *53 Malvern road* may have 3 units, with subaddresses *unit 1, unit 2 & unit 3.*

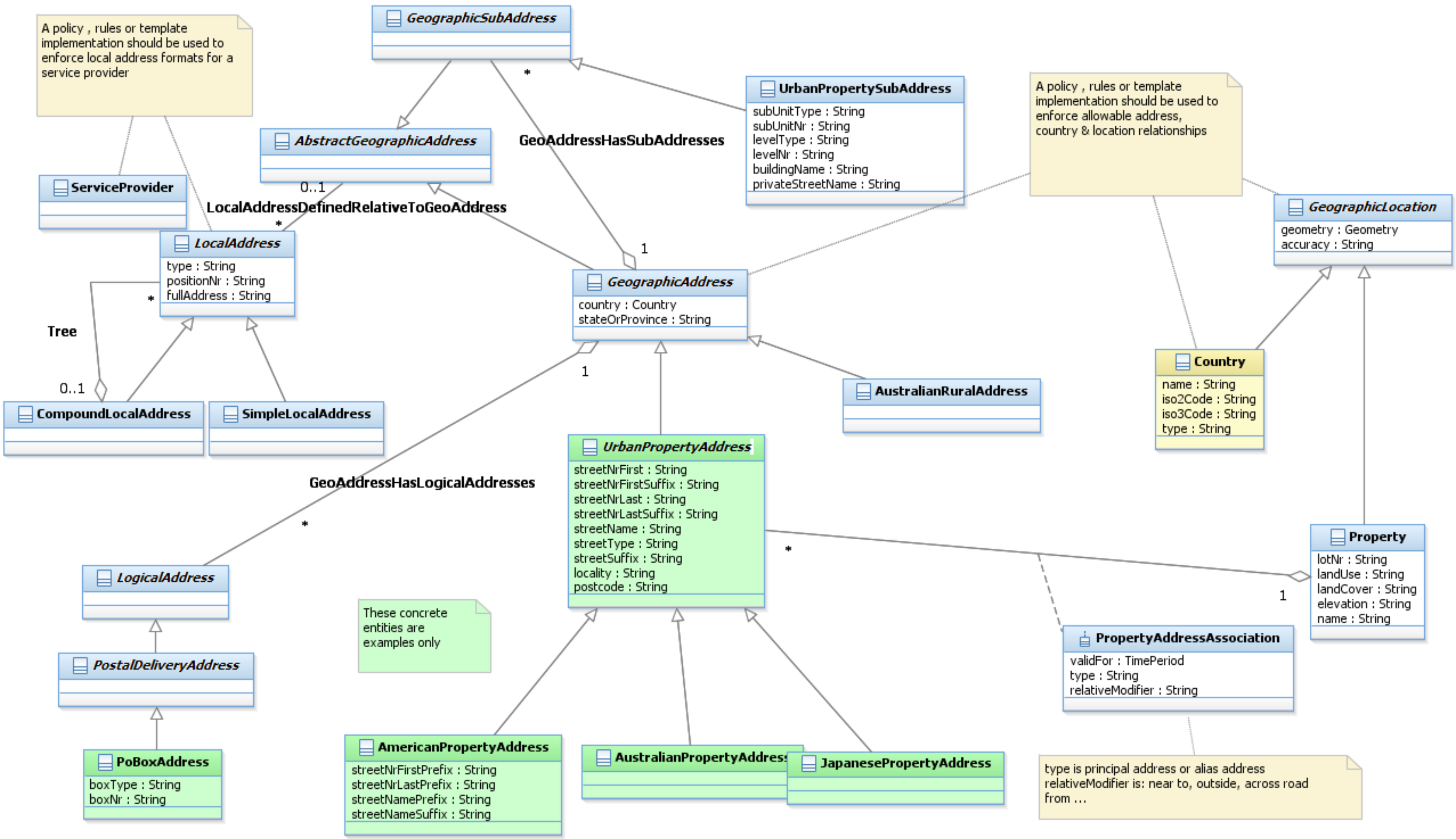



**Figure L.12 – Types of Address**

Note that in figure 12 that a full set of Addresses has not been shown. The example entities have been shown for example only, on how the core model can be extended in particular situations.

## Types of Local Address

A Local address could be of the form: Building 1 / Floor 6 / Suite 25 / rack 12 / subrack 1 / card 7

The Local Address Policy should enforce Service Provider specific Local address business rules.

A possible naming graph is shown in fig 14. This would form the basis of a naming policy.

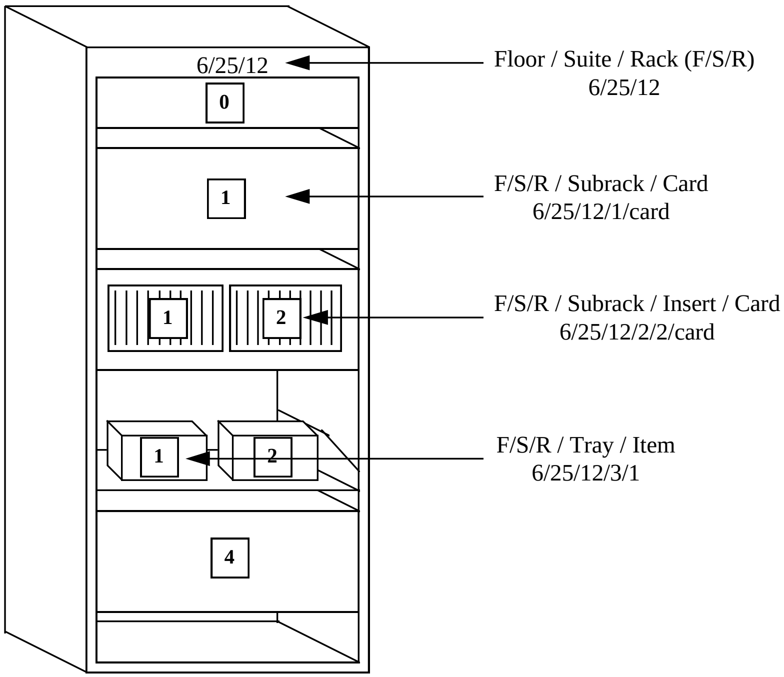
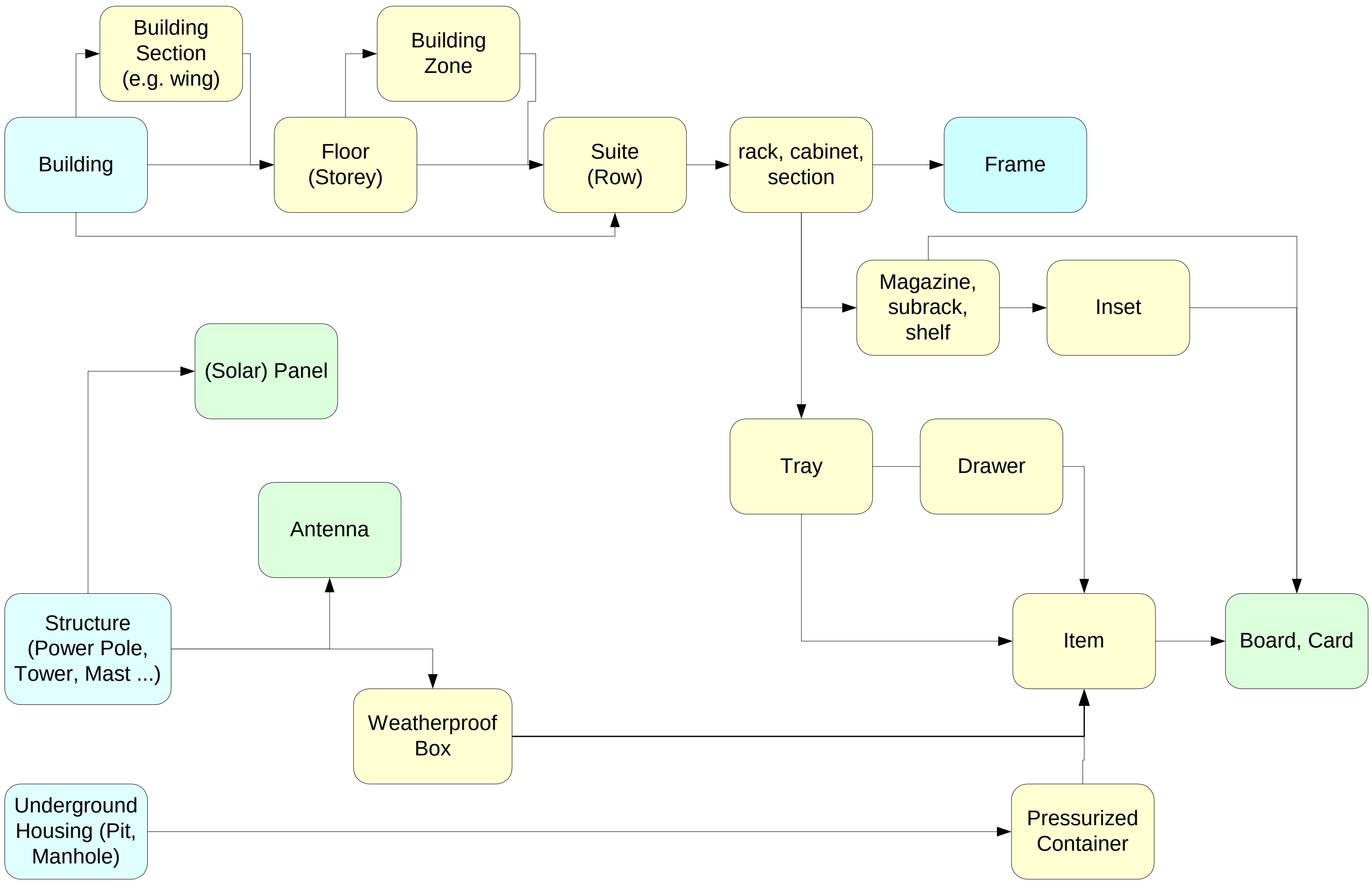



**Figure L.13 – Example of a Rack**




Note: This is NOT a UML diagram

**Figure L.14 – Example of a Network Inventory Naming Graph**

## Location & Address Role

In the Party SID model facet, we defined roles that parties (people and organizations) may play. In a similar fashion, we may find it useful to define location and address roles. This will allow us to use a single location or address for different purposes, e.g. a single address may be a:

- Customer billing address
- Phone directory address
- CPE equipment location address and a
- Geographic plant reference address (pillar outside the front of …).

Because the role requirements are simpler than in the Party SID model facet, Location roles have been modeled as associations. [Fowler-role]

Care should be taken not to confuse when to create address & location subtypes versus address & location roles.

## Interworking with the rest of the SID model

Fig 15 shows some possible connections from Location to other parts of the SID model. This diagram is intended to be illustrative rather than prescriptive.

The SID model doesn't store GeoSpatial information as a part of every entity. As each entity is modeled, we determine whether the entity *has* GeoSpatial attributes or *refers to* GeoSpatial attributes. For network equipment, we create abstract GeoSpatial entities of location and route section etc. and reference these from the network elements.

To reiterate; Location information should not be stored in Product, Party or Network Resource entities but associated with a Place.

If a Place exists, it should be shared and not duplicated.

This allows queries that would otherwise require GeoSpatial queries to be done textually, i.e. to check if two network elements are at the same location, just see if the place ids are the same.

When placing a Resource or a Product, the choice is to associate to an:

- address, or
- site, or
- location

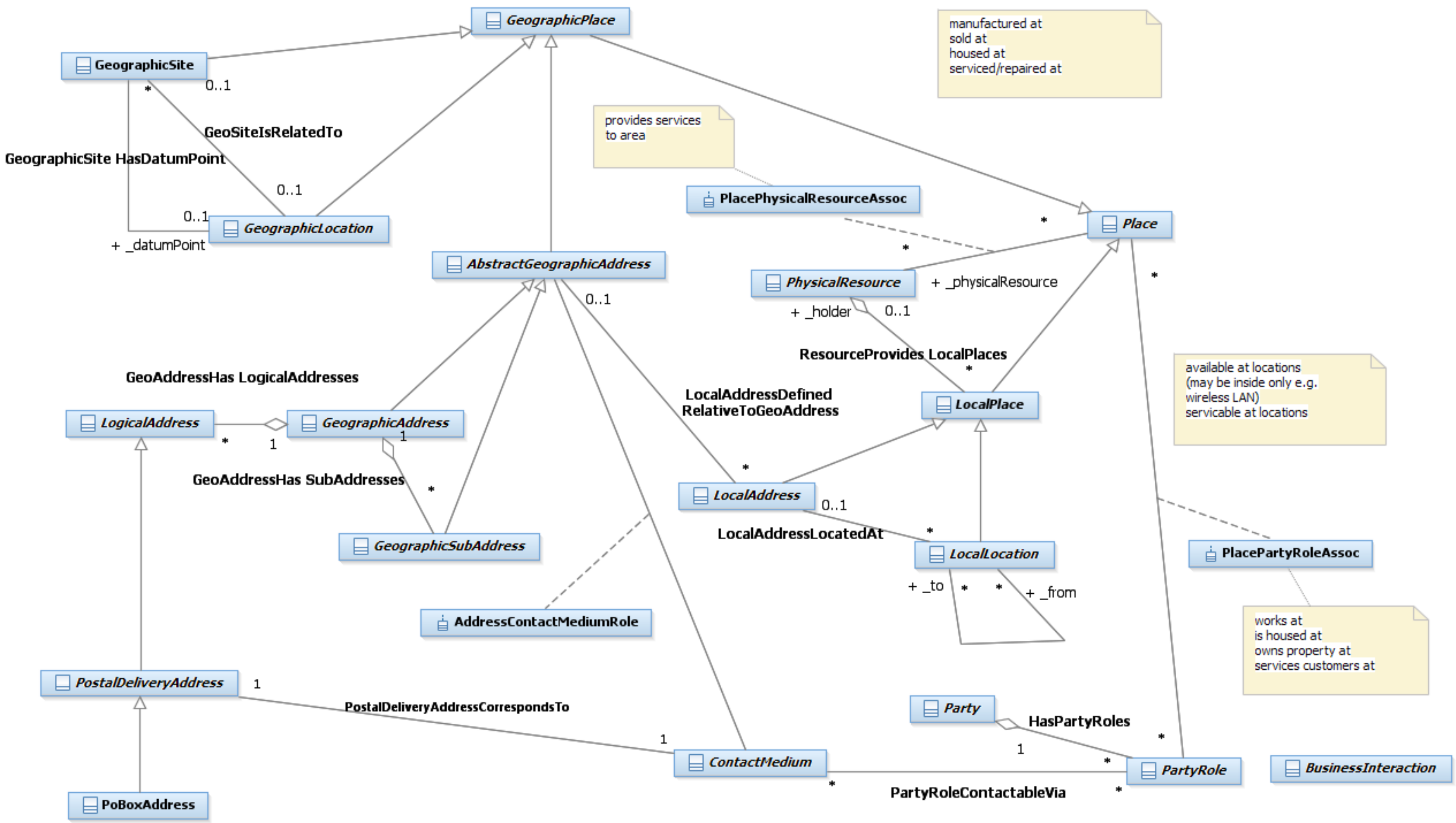



**Figure L.15 – Links to Other Parts of the SID Model**

## Using this model for Process Analysis

When using these entity definitions in process or use case analysis, the most useful entities will be

- Geographic Address
- Geographic sub address
- Postal Delivery Address
- Geographic Site & Role
- and the concrete example Geographic Locations (Street, Property, Country, State …)

If a generic term for "where?" is required and the answer could be an Address a defined Site or a Location, then the term Place should be used.

In common language the Geographic / Local prefix is usually not used as it is assumed that it is clear from the context. During process analysis it is probably worthwhile to explicitly note this information.

It will also be useful to note whether an address is a Postal Address or not.

If more detailed entities are referenced, then they should be defined and related back to the entities in this model. e.g. "a Council Area is a type of Administrative Area that …"

## Model Implementation Using Components

This model should not be implemented as a monolithic component, but in such a way that functionality can be plugged in as required.

This is shown conceptually in fig 16

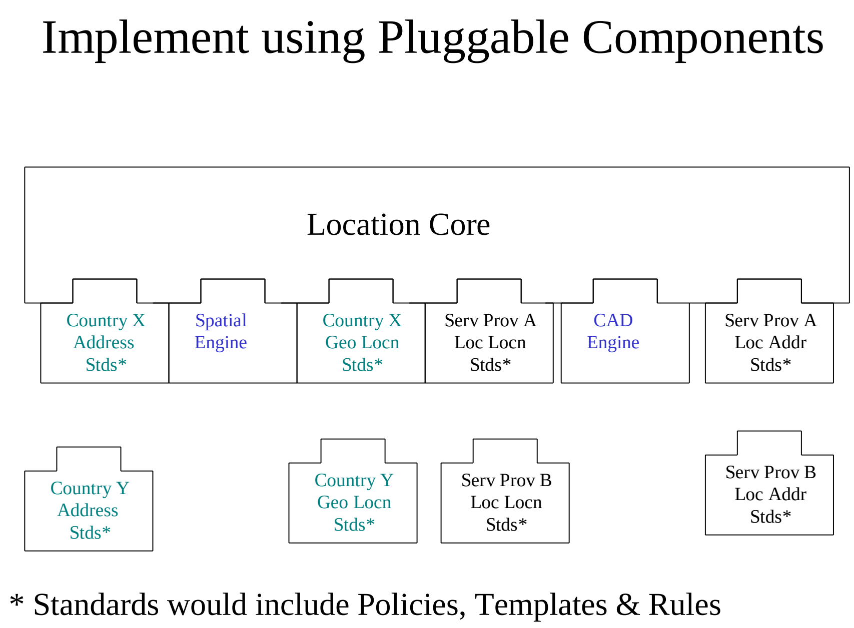



**Figure L.16 - Implement using Pluggable Components**

Note that a Service Provider may need to have multiple standard sets plugged in at the one time.

## Further Work Required in Later Releases

This section documents some issues that have not been resolved in this version.

- Need to ensure that we have a clear interface with the Physical Resource model. As the Location and Physical Resource models are being defined in parallel, this interface will be refined with successive model versions. In particular, if we should allow arbitrary local place definitions by the definition of a reference point, not requiring any PhysicalResource to be present.

- Need to investigate the use of IAI / IFC 2.x model for Local places [IAI/IFC]

- Should try to make concrete location entities compliant with Spatial Data Standard SDSFIE/FMSFIE Release 2.10 [SDSFIE/FMSFIE]

- Need to validate how an equipment suite (row) should be handled, should we make it a new local coordinate system, or an absolute Local place, that fixes one of the x or y coordinates and allows placement relative to it?

- The current model has some overlap between Geographic sub address and Local address, we need to think more about this (refer fig 5). Local Address should probably only be defined where Geographic address finishes. This means that Local Address will not have a standard starting point.

- Investigate the removal of the Street and Network Route specializations to Geographic Place.

- Re-look at Site Role and how this is expected to be used

- Should add CAD discussions to “Graphical or Textual“ section

- Need to look at Local Location again to see if it can be made less dependent on EquipmentHolder. This will also fit in with local Place definitions for sites. Also we will need Graphical Representations for Local Places like suite, which aren’t defined by Physical Resource.

- Determine if a limited Composite pattern should be used for Symbology (ie. add simple and compound symbology entities to fig. 10b) to allow for GeographicLocation types with multiple Geometry attributes.

- Consider if LogicalAddress should link into GeographicAddress, AbstractGeographicAddress or higher up un the hierarchy

What about things like a technician’s van containing card spares?? Should we model these??

- we could model a van as a holder with a Local location. We may not know where the van is, but just know that the item is "in the van". Some service providers may keep track of a van location and be able to send the closest van with the required capability to fix a fault or install a service.

## Notes:

- Note that these entities represent business concepts in a Service Provider that conforms to the eTOM process model [eTOM].
- Entities that are outside the scope of this model facet are shown with a white fill color.
- This is intended as a “minimalist” model. Subtypes and attributes should be added as required. The attributes shown should be considered as suggested, not required.
- Parent attributes are not repeated in the sub entities
- In most cases relationships are not documented as attributes
- Only significant entities are shown in the “related business entities” cells

## Standard citation for this document

TM Forum SID GB922 Addendum 1L (Location)

Chris Hartley, Helen Hepburn, et al.

URL: [http://www.tmforum.org/](http://www.tmforum.org/)

## References


| | |
|---|---|
| Fowler-AP | Analysis Patterns – Reusable Object Models<br>by Martin Fowler ISBN 0-201-89542-0<br>http://martinfowler.com/articles.html |
| Fowler-Role | Dealing with Roles<br>http://www.awl.com/cseng/titles/0-201-89542-0/apsupp/roles2-1.html |
| Fowler-Time | Patterns for things that change with time<br>http://martinfowler.com/ap2/timeNarrative.html |
| ACIA | Alliance Common Information Architecture (AT&T, BT & Concert) version 3.1, 15 Dec 2000 |
| eTOM | GB921 version 2.5 |
| OGIS | Open GIS Consortium<br>http://www.opengis.org/ |
| GML | Geographic Markup Language V2.0<br>http://opengis.net/gml/01-029/GML2.html<br><br>Introduction to Geography Markup Language (GML)<br>http://www.jlocationservices.com/company/galdos/articles/introduction_to_gml.htm |
| Zachman | Zachman Framework<br>http://www.zifa.com/frmwork2.htm |
| Coad-Roles | Coad Letter 76 – January 2001<br>http://www.togethercommunity.com/coad-letter/Coad-Letter-0076.html |
| Coad-Archetypes | Modeling with Color<br>http://www.togethersoft.com/services/tutorials/jmcu/index.html |
| ANZLIC-STREET | draft Australian and New Zealand Rural and Urban Addressing Standard<br>http://www.anzlic.org.au/icsm/street/data_model.pdf |
| ADDRESS-XML | Postal Address Version 1.1<br>http://www.xml.org/ |
| ISO-3166 | ISO 3166 Country Codes (2 Character)<br>http://www.din.de/gremien/nas/nabd/iso3166ma/ |
| COUNTRY-CODES | Countries or areas, codes and abbreviations (3 char)<br>http://www.un.org/Depts/unsd/methods/m49alpha.htm |
| Bus-Patterns | Business Modeling with UML, Business Patterns at Work<br>Hans-Erik Eriksson, Magnus Penker ISBN 0-471-29551-5 |
| M3100 | Generic network information model – Recommendation M.3100 (07/95)<br>http://www.itu.int/rec/recommendation.asp?type=folders&lang=e&parent=T-REC-M.3100 |
| AS4590 | Australian Standard AS 4590-1999 : Interchange of client information |

| | |
|---|---|
| | http://www.standards.com.au/catalogue/script/Details.asp?DocN=stds000023947 |
| Baumer | The Role Object (Design) Pattern. Download PDF from http://www.riehle.org/computer-science-research/1997/plop-1997-role-object.html |
| Hay | Data Model Patterns , Conventions of Thought http://www.dorsethouse.com/books/dmp.html |
| Cadastre | Commission 7 Working Group 1 "Reforming the Cadastre" http://www.swisstopo.ch/fig-wg71/cad2014.htm Cadastre 2014 – A VISION FOR A FUTURE CADASTRAL SYSTEM http://www.swisstopo.ch/fig-wg71/cad2014/download/cad2014_eng.pdf |
| Oracle Spatial Concepts | Oracle Spatial User's Guide and Reference Release 9.2 http://download-west.oracle.com/docs/cd/B10501_01/appdev.920/a96630.pdf |
| OLS | Open Location Services Initiative http://www.openls.org/ |
| AP ADDR STDS | Australia Post – Presentation Standards Booklet. http://www.auspost.com.au/futurepost/index.asp?link_id=9.1575 |
| LAND USE | US Geological Survey – LAND USE/LAND COVER CLASSIFICATION –Modified Anderson Classification http://rockyweb.cr.usgs.gov/public/mrgb/mrgblulcdefs.html |
| Clementini | E. Clementini, P. Di Felice, and P. van Oosterom. A Small Set of Formal Topological Relationships for End-User Interaction http://dsiaq1.ing.univaq.it/hpage/eliseo/research.htm – s1 |
| Egenhofer | Max J. Egenhofer's List of Publications http://www.spatial.maine.edu/~max/pubs.html |
| ISO / DIS 19109 | Draft International Standard. Geographic Information – Rules for Application Schema |
| ANSI T1.253-1999 | American National Standard for Telecommunications - Information Interchange - Code Description and Codes for the Identification of Location Entities for the North American Telecommunications System |
| USPS-28 | Postal Addressing Standards - Publication 28, November 2000 http://pe.usps.gov/cpim/ftp/pubs/Pub28/pub28.pdf |
| Land Use Database | The Land Use Database Manual http://www.itc.nl/education/larus/landuse/landuse.html |
| SDSFIE/FMSFIE | Spatial Data Standard SDSFIE/FMSFIE Release 2.10 : (ANSI Standard NCITS 353) http://tsc.wes.army.mil/products/tssds-tsfms/tssds/sdsnews.asp |
| IAI/IFC | International Alliance for Interoperability (IAI). http://iaiweb.lbl.gov/ |
| ISO 10303 | ISO 10303 : Industrial automation systems and integration Around 60 parts have been published, many of these are of interest for Location |
| Composite Pattern | Articles: Composite à la Java, Part I & II http://www.research.ibm.com/designpatterns/publications.htm |
| JTS | Java Topology Suite http://www.vividsolutions.com/jts/jtshome.htm |

| | |
|---|---|
| ESRI Models | ESRI Data Models<br>http://support.esri.com/index.cfm?fa=downloads.dataModels.gateway |
| Calgary Model | An ArcGIS Address Data Model for the City of Calgary<br>http://downloads.esri.com/support/datamodels/Address/An_ArcGIS_Address_Data_Model_for_the_City_of_Calgary.pdf |

## Examples

### Example 1 – A place increasing in complexity over time

In this example we will look at a simple Place that changes over time and how the model allows for these changes.

At the start we have a simple church that has two POTS lines. The Service Provider initially just records the Geographic Location of the property boundary and the address of the property.

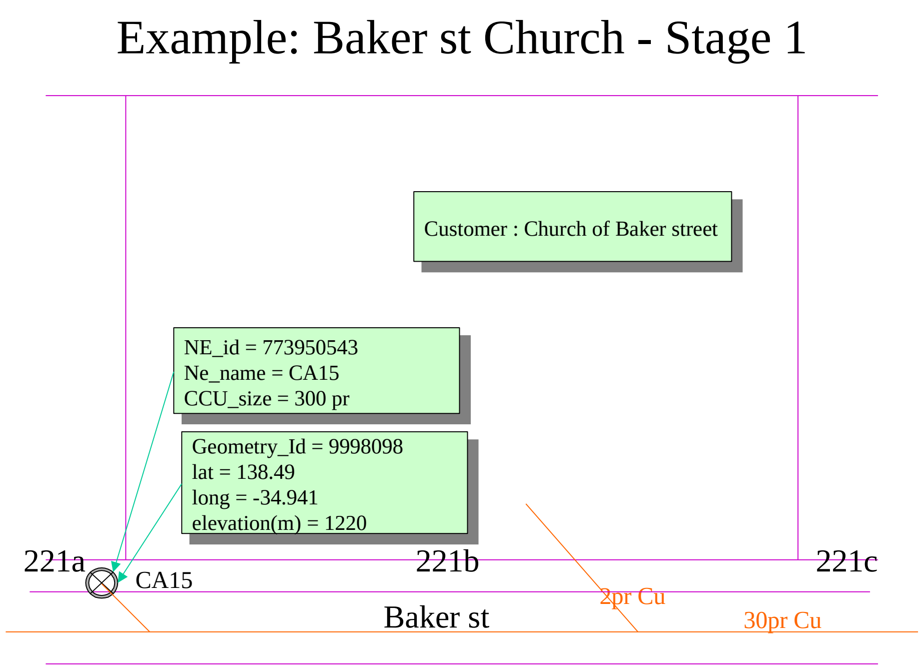



**Figure E.1.1 – A Simple Place**

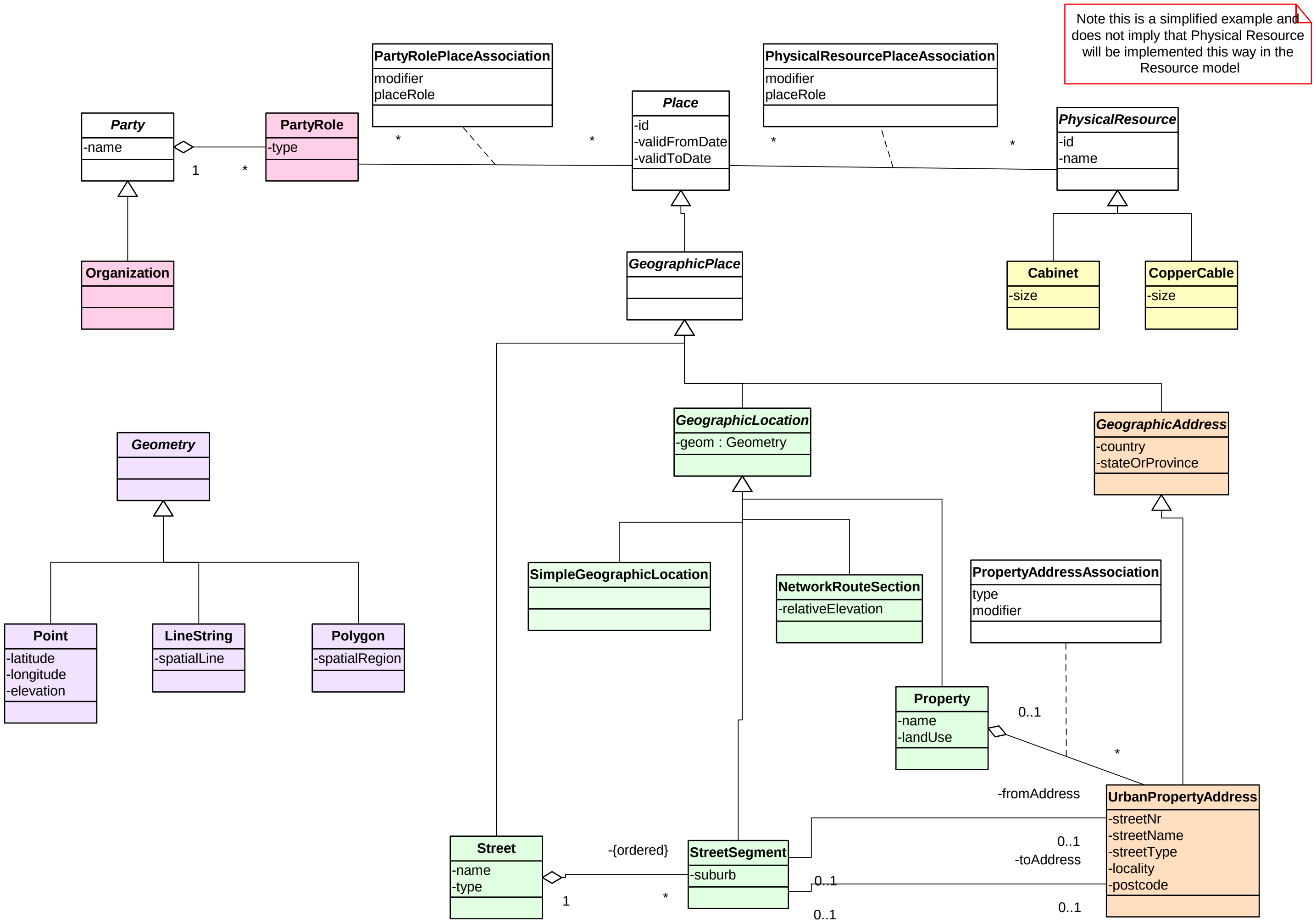



**Figure E.1.2 – Class Diagram 1**

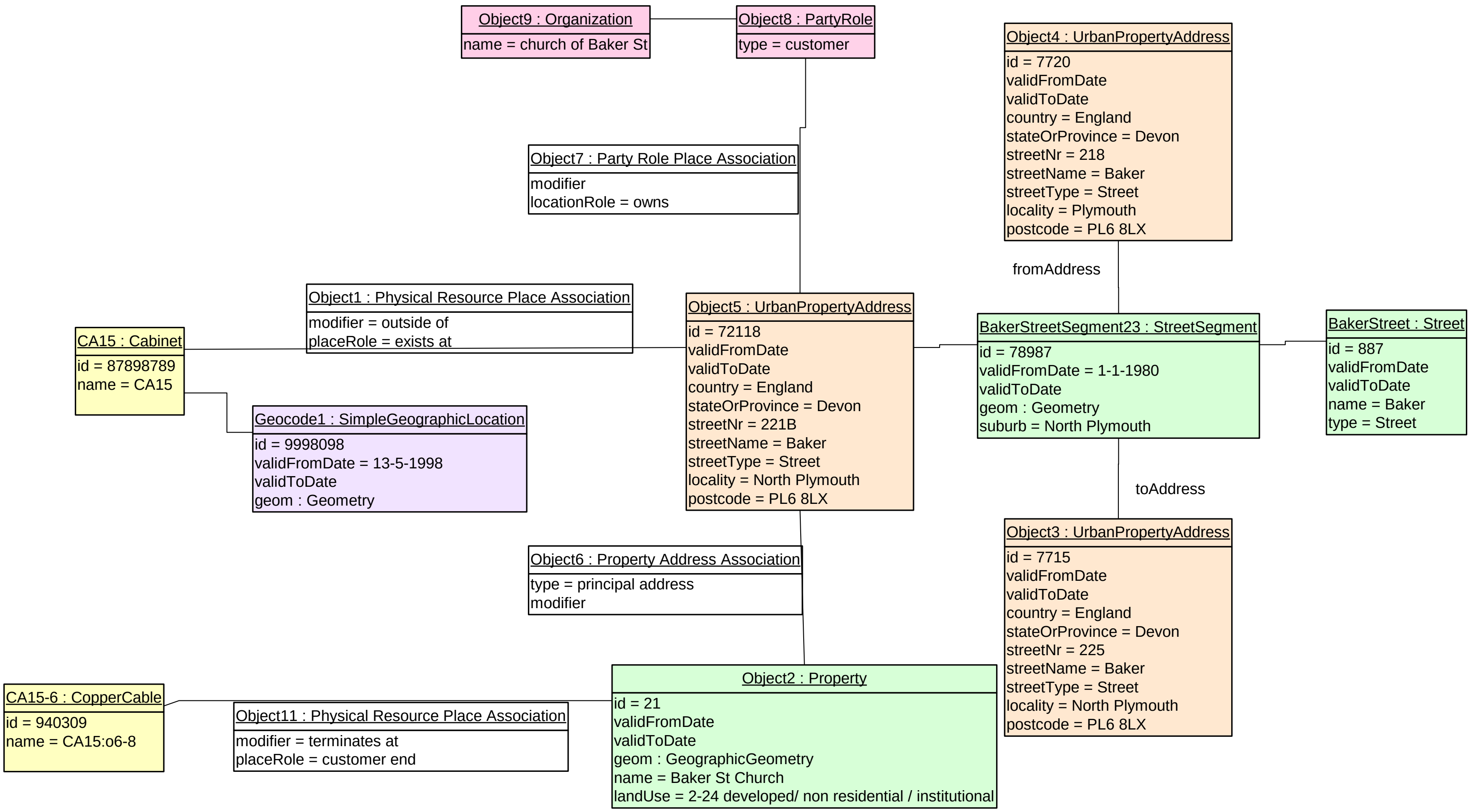



**Figure E.1.3 – Object Diagram 1**

The church builds a second (administration) building and comes to an agreement to lease use of the steeple space to house a mobile phone antenna. Mobile base station equipment is housed in a room of the new administration building. The service provider produces CAD drawings of the Site and their equipment room and stores additional information about site access, contacts, keys …

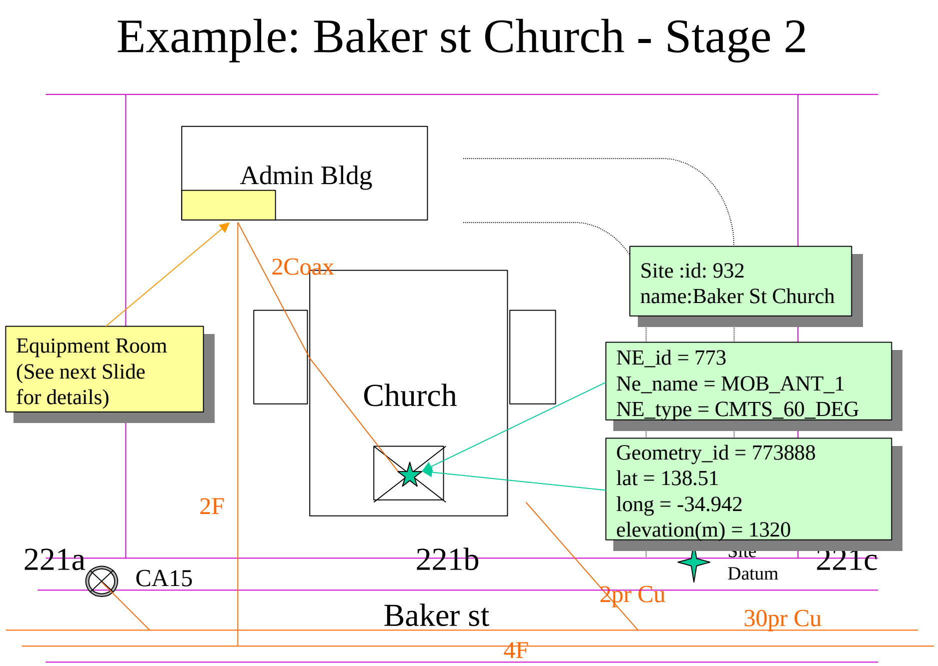



**Figure E.1.4 – A Simple Site**

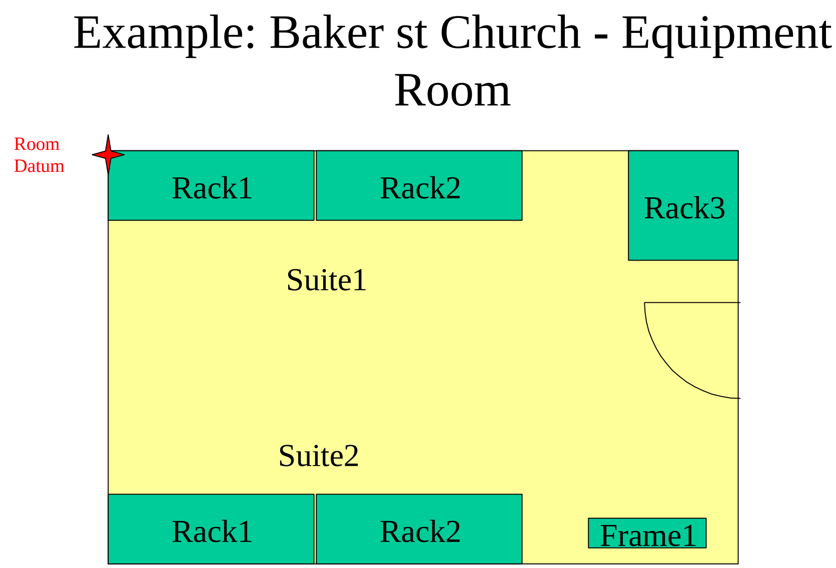



**Figure E.1.5 – A Simple Local Place**

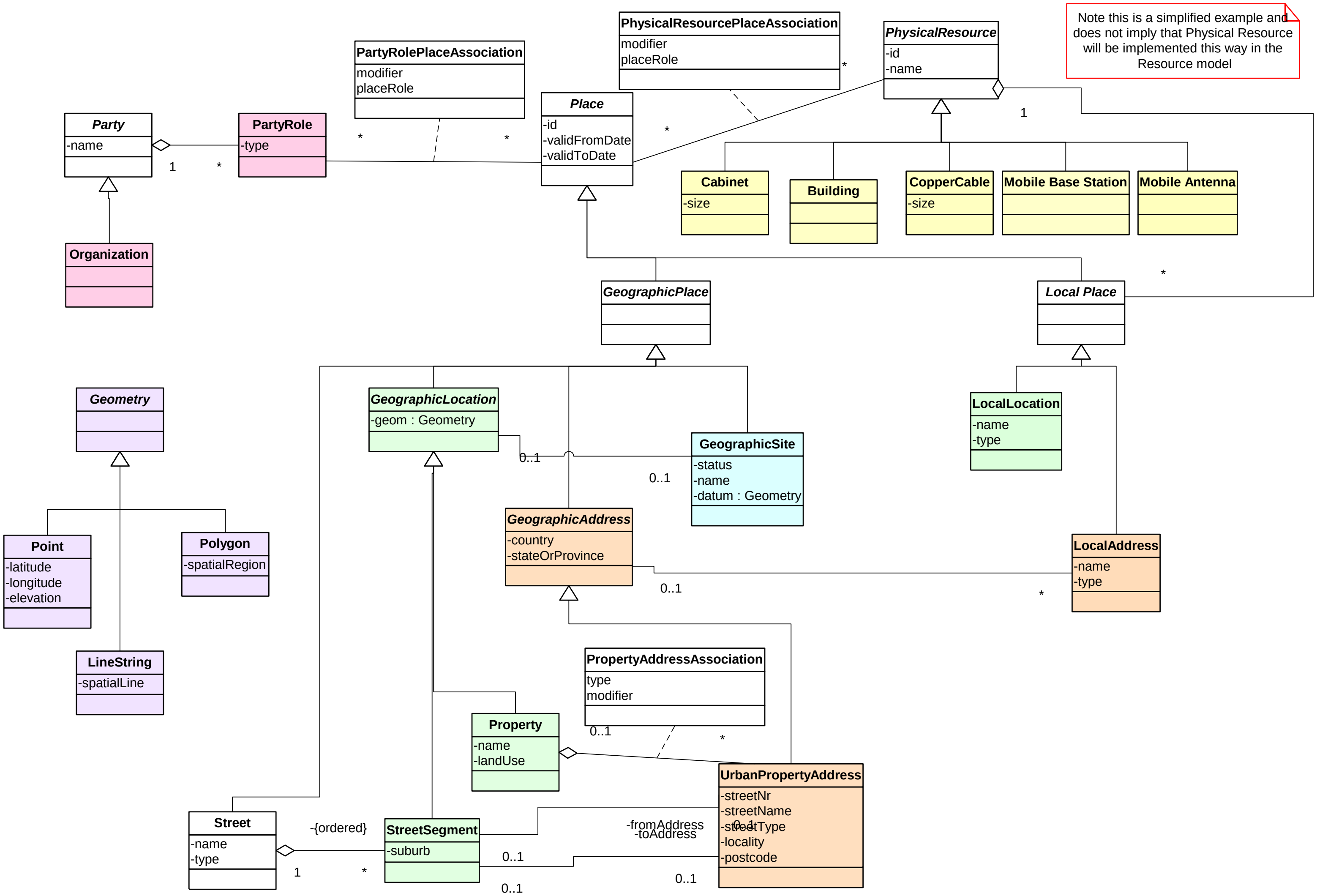



**Figure E.1.6 – Class Diagram 2**

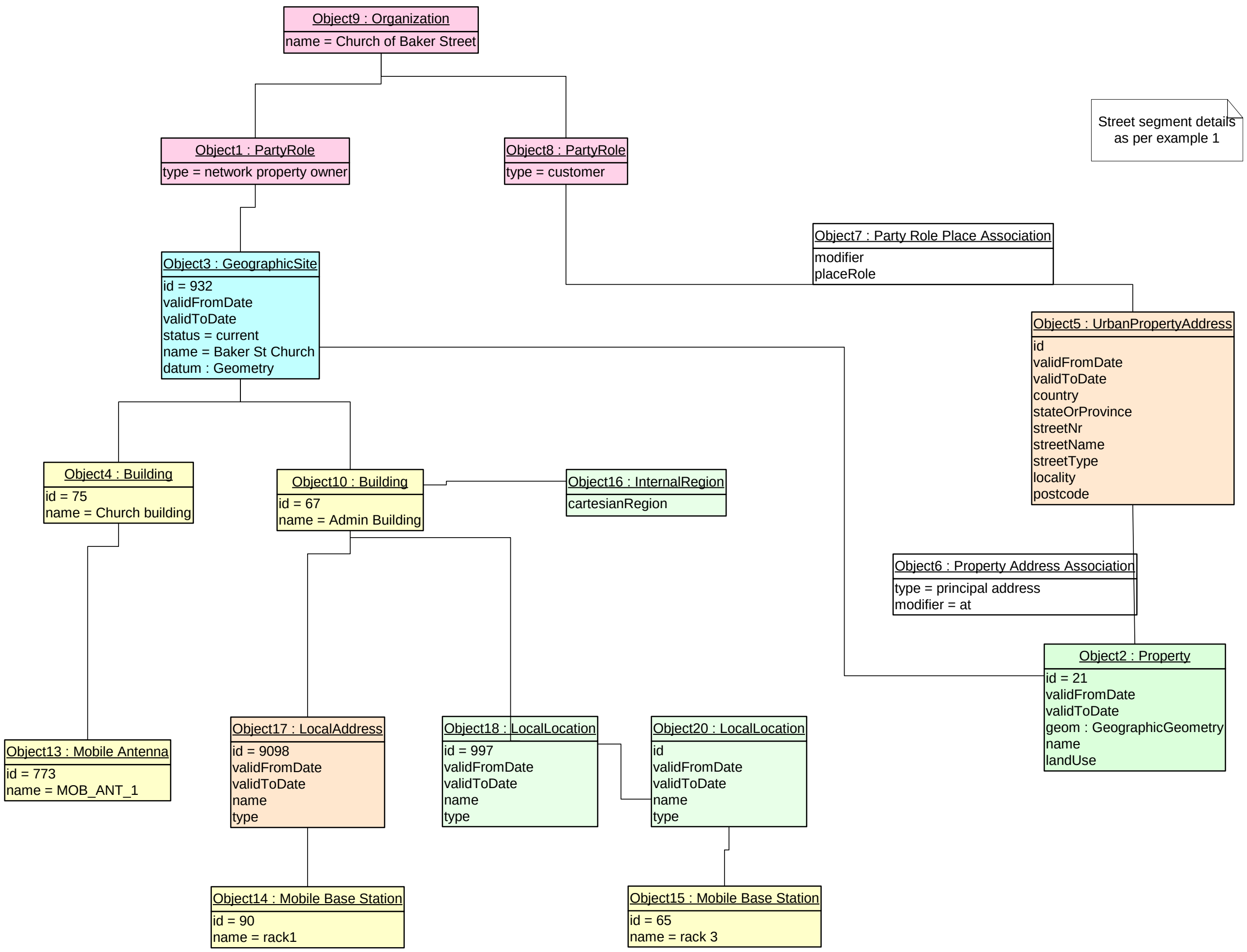



**Figure E.1.7 – Object Diagram 2**

### Example 2 – A Cable Route

This is a simple example showing how Network Elements should be separated from their geographic attributes.

Route sections are defined with geographic attributes and can be shared by any Resources on that route.

Routes are defined as an ordered collection of route sections. As defined here, routes have no geographical attributes of their own.

Cables are associated to the Route. In this example our copper cable takes route AB1.

Note that other cables (not shown) may share a route.

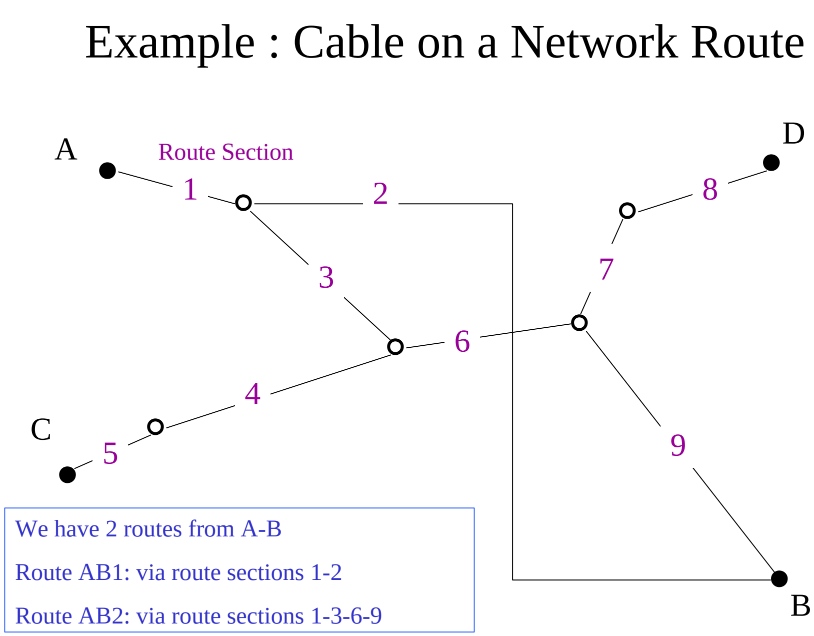


**Figure E.2.1 – Network Diagram**

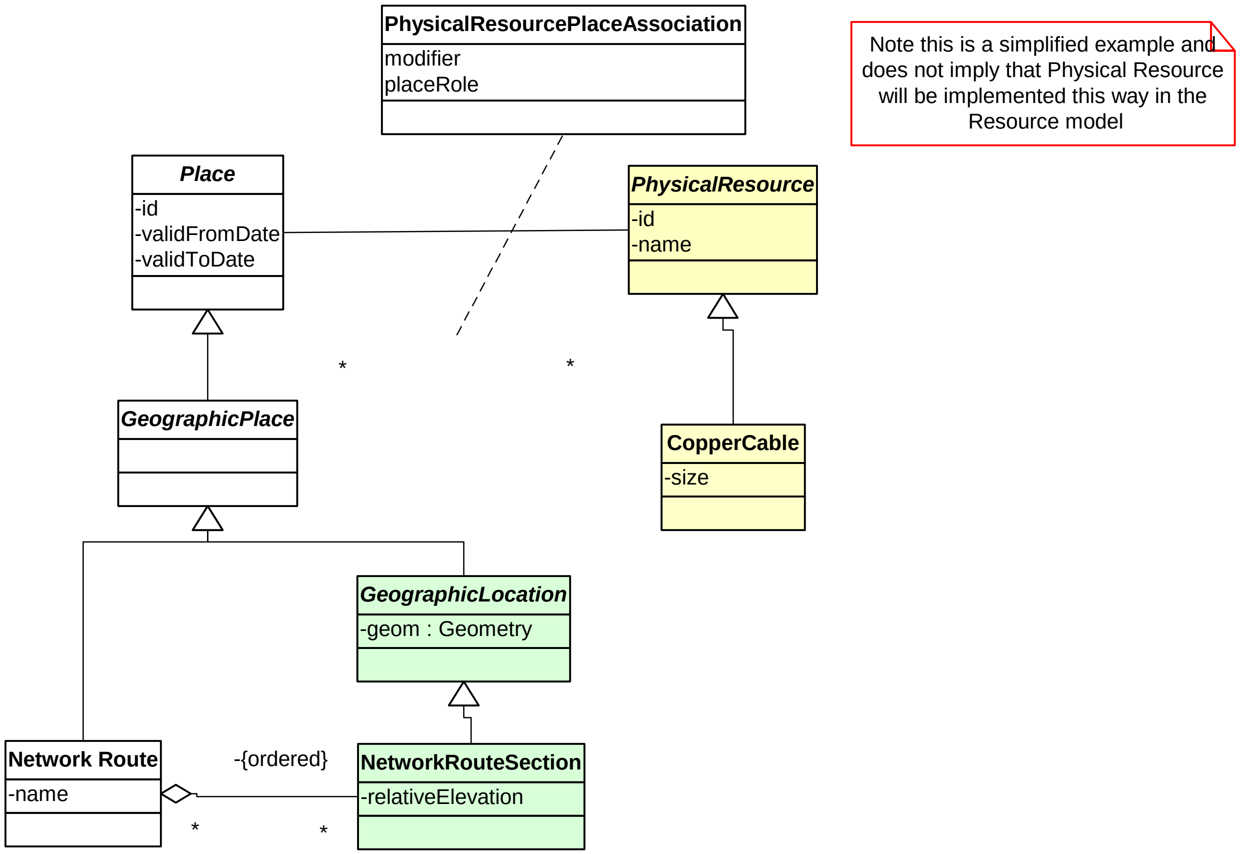



**Figure E.2.2 - Cable Route Class Diagram**

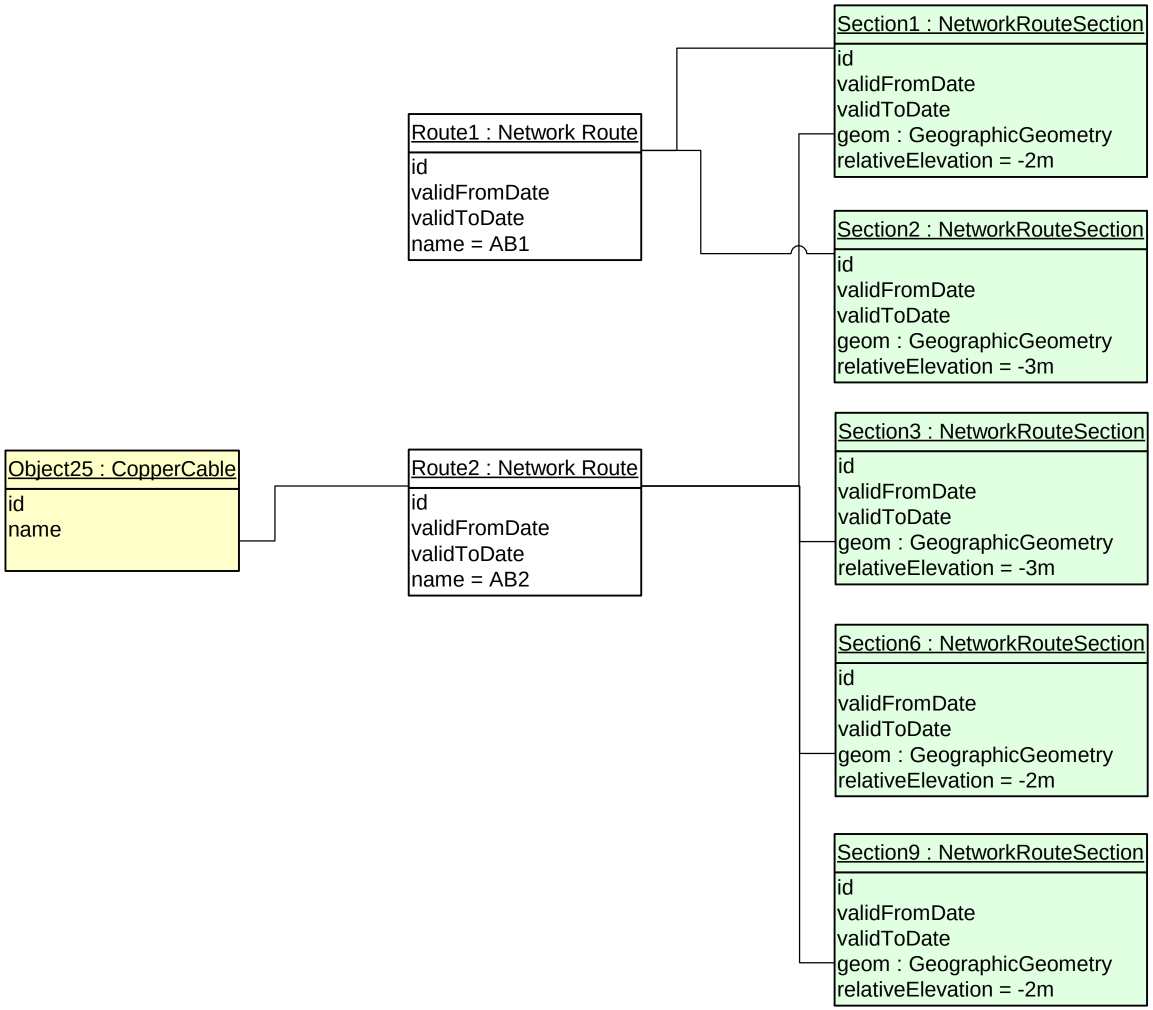



**Figure E.2.3 - Cable Route Object Diagram**

### Example 3 – A Complex Address

<<This example will be provided in a later version>>

### Example 4 – A Multi Story / Multi Tenanted Building

"Green & Gold" house is a multistory building in the central business district. This Service Provider has some human resources staff on the 5th floor.

Janet Green is employed in the "Vitality Enablement" section of HR.

On the 3rd floor there is a consulting group, which is a customer of this Service Provider. One of the Service Provider's Products terminates there. If the place was deemed important enough, then a site & building entity could also be created.

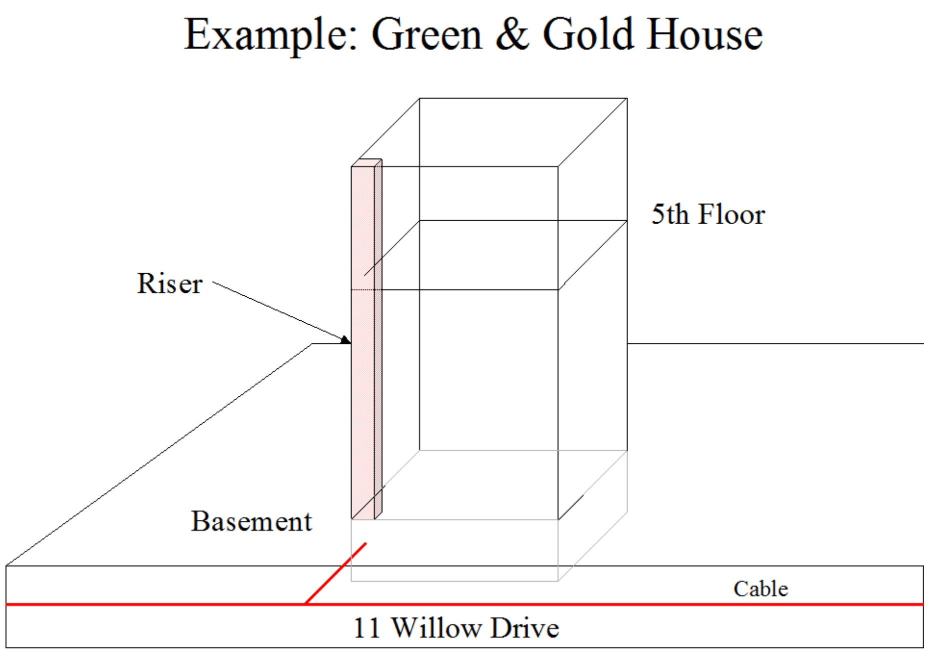
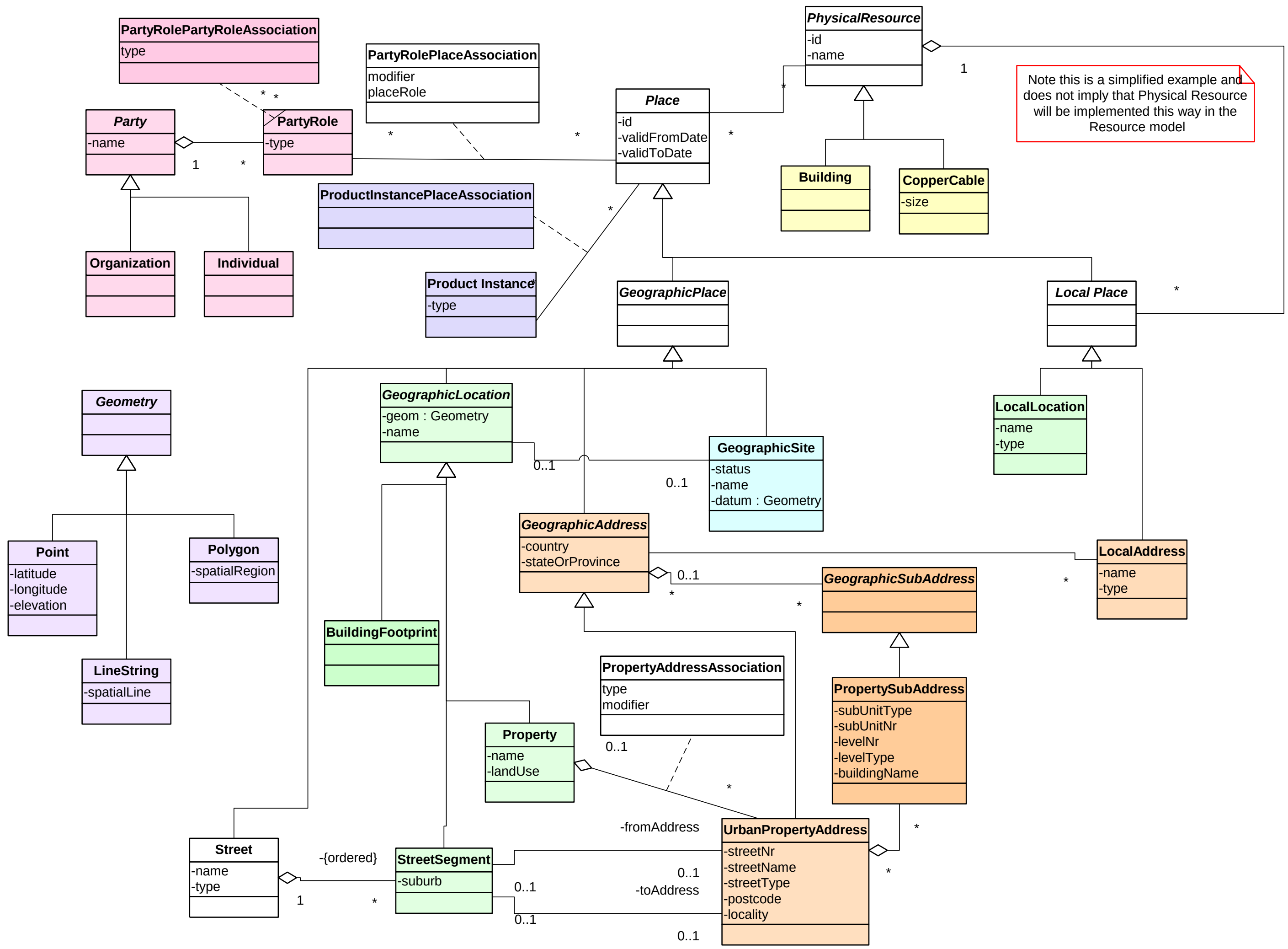



**Figure E.4.1 - Multistory Building**




## Figure E.4.2 - Multistory Building Class Diagram

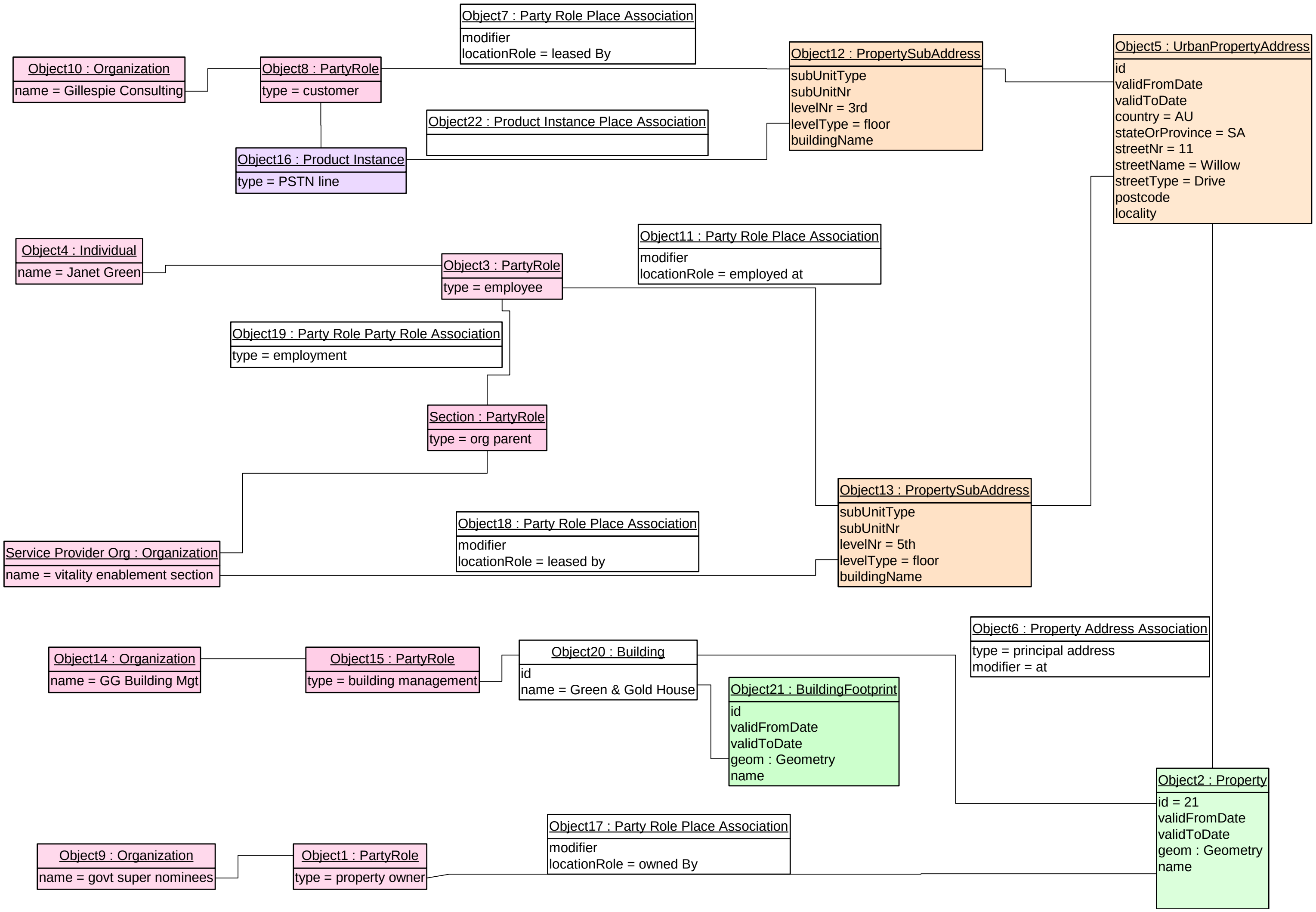



**Figure E.4.3 - Multistory Building Object Diagram**

## Example 5 – Location based Service Support

This example shows how the SID Location model can support location based services.

A customer rings up the “Find the Nearest Pizza Barn” number using her mobile phone.

The mobile phone network determines the approximate position of the calling phone.

Using a spatial engine, the two closest “Pizza Barn” restaurants are calculated.

This information can now be returned to the caller as a map, as text or using synthesized voice.

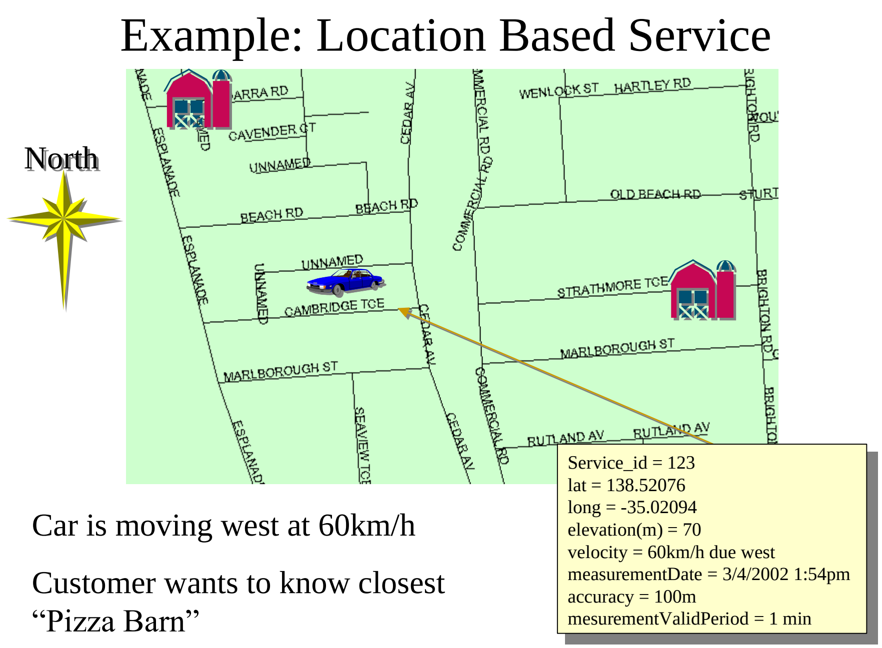



**Figure E.5.1 - Nearest “Pizza Barn”**

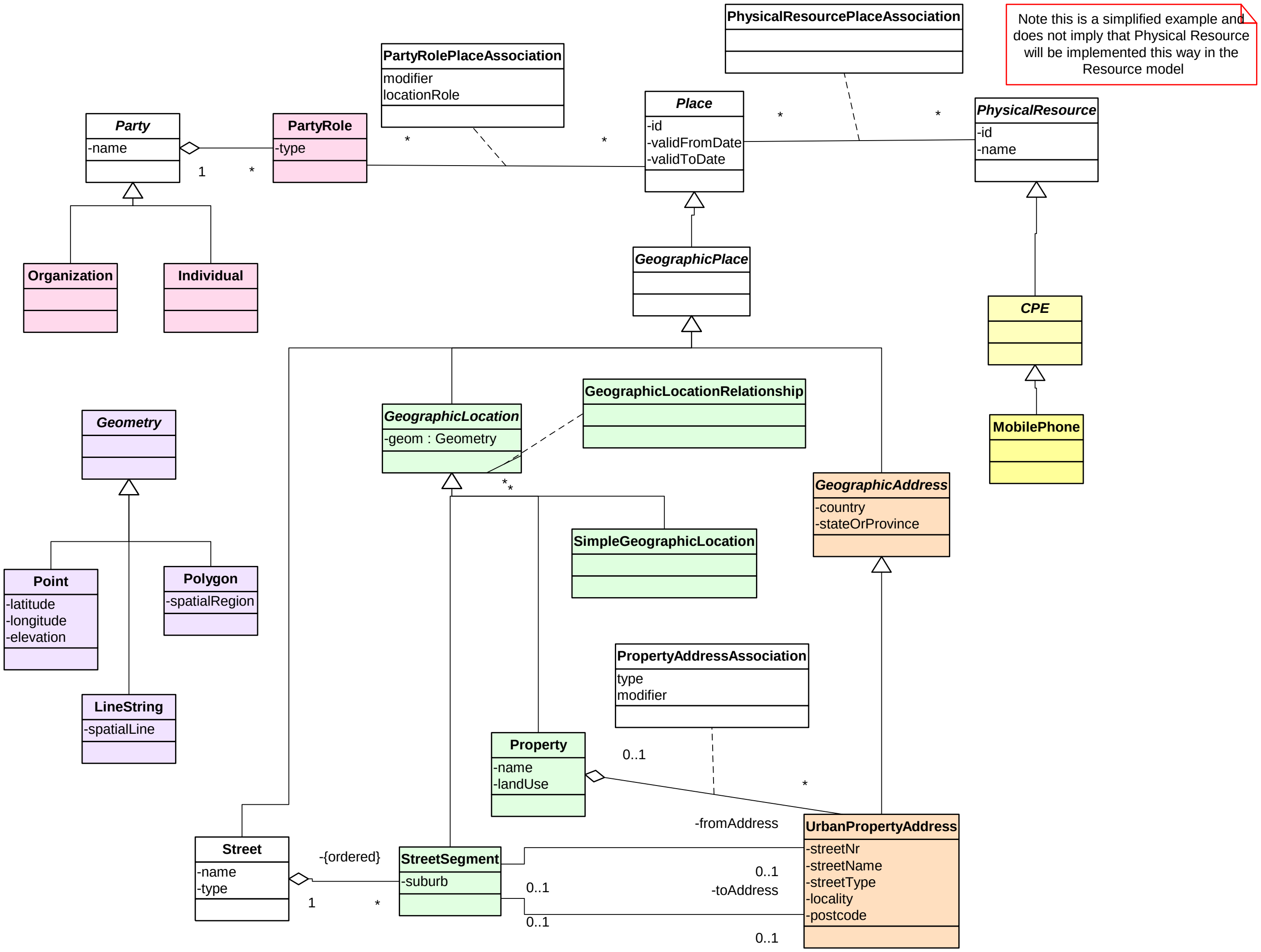



**Figure E.5.2 - Location Based Service Class Diagram**

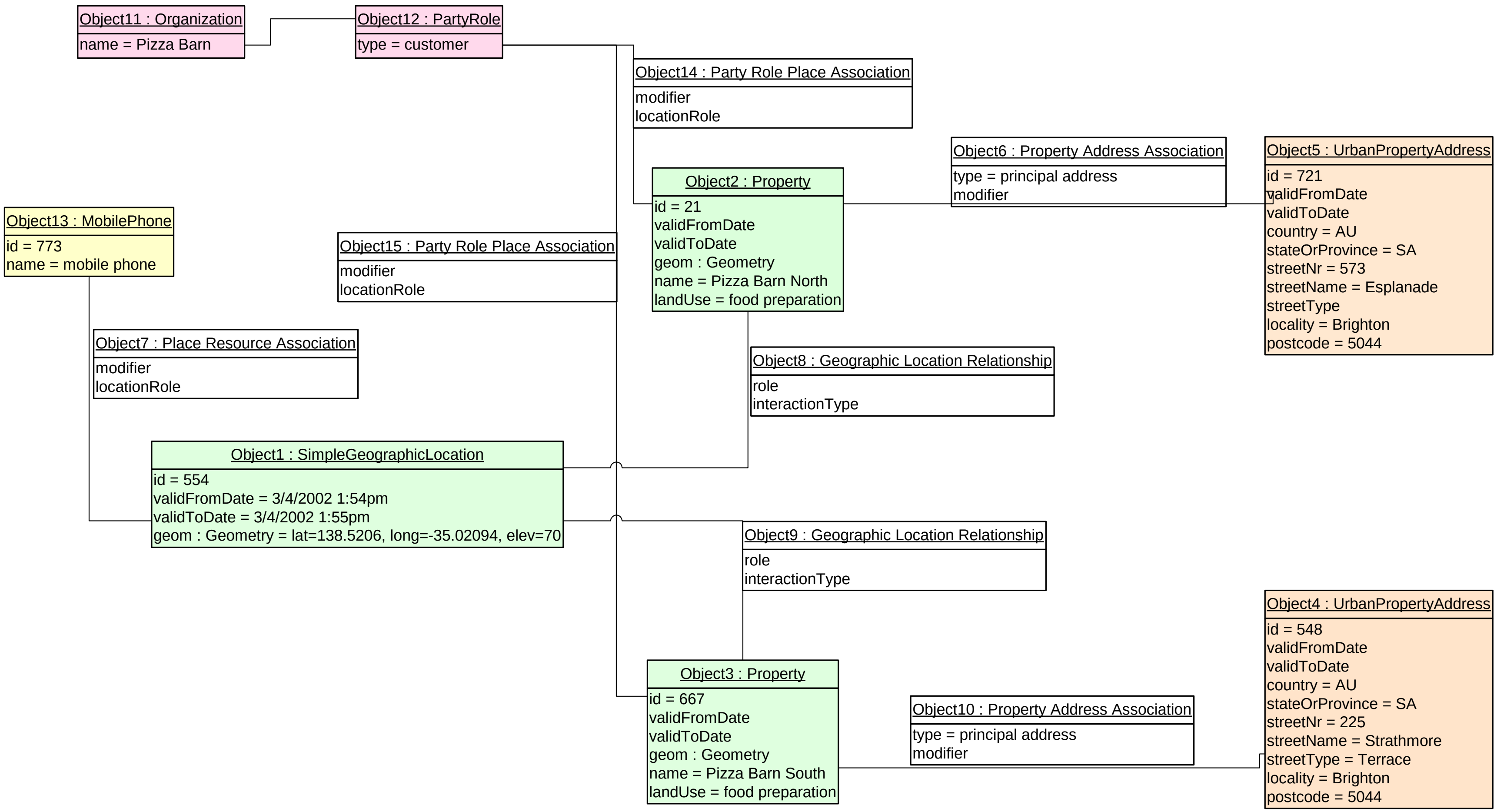



**Figure E.5.3 - Location Based Service Object Diagram**

# About this document

This is a TM Forum Guidebook. The guidebook format is used when:

- The document lays out a 'core' part of TM Forum's approach to automating business processes. Such guidebooks would include the Telecom Operations Map and the Technology Integration Map, but not the detailed specifications that are developed in support of the approach.
- Information about TM Forum policy, or goals or programs is provided, such as the Strategic Plan or Operating Plan.
- Information about the marketplace is provided, as in the report on the size of the OSS market.

## Document History

| Version | Date | Purpose |
|---|---|---|
| 0.1 | Jan 2002 | Document created. Location copied from old template. |
| 0.2d | Apr 2002 | Updated for TMW Nice |
| 0.5b | Apr 2002 | Final TMW Nice Version |
| 3.0 | Jun 2003 | Updated for phase 3 release |
| 3.1 | July 2003 | Formatted for Member Review with NGOSS R3.5 |
| 3.2 | Aug 2004 | To reflect TMF Approval Status |
| 3.3 | Jan 2011 | Update Notice |
| 3.4 | Mar 2011 | Updated Notice, minor formatting corrections prior to web posting and ME |
| 3.5 | Sep 2011 | Updated to reflect TM Forum Approved status |
| 3.6 | Jun 2013 | Integration of CR Artf1431 (Figures 3, 8, 9, 11d, 12) |
| 3.7 | Jul 2013 | Updated cover & header, corrected notice & footer prior to posting |
| 3.8 | Sep 2013 | Added IPR Mode to cover, updated notice to reflect TM Forum Approved status |
| 3.8.1 | Nov 2013 | Updated cover, header, footer & Notice |
| 4.0 | Apr 2014 | Copy of all pertinent information to the model and removal of Business Entity Definition section |
| 4.0.1 | May 2014 | Updated Notice & footer, moved Admin section to end; minor formatting edits prior to posting |
| 14.5.0 | Oct 2014 | Updated figure numbers and figures. |
| 14.5.1 | Mar 2015 | Updated cover, header, footer and Notice to reflect TM Forum Approved status |

# Acknowledgments

This document was prepared by the members of the TM Forum Information Framework (SID) team:

The Shared Information/Data Model is a genuinely collaborative effort. The TM Forum would like to thank the following people for contributing their time and expertise to the production of this document. It is just not possible to recognize all the organizations and individuals that have contributed or influenced the introduction. We apologize to any person or organization we inadvertently missed in these acknowledgments.

Key individuals that reviewed, provided input, managed, and determined how to utilize inputs coming from all over the world, and really made this document happen were:

| Name | Affiliation |
|---|---|
| Cliff Faurer | TM Forum |
| Chris Hartley | Telstra |
| Helen Hepburn | British Telecom |
| Dan Matheson | Hewlett-Packard |
| John Reilly | MetaSolv Software |
| Wayne Sigley | Telstra |
| John Strassner | Intelliden |
| Johan Vandenberghe | Lucent Technologies |